

Standard Software & Visualization Tools

Lecture 3

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Gaussian & GaussView

PRO	CONTRA
Excellent internal coordinate module supporting very sophisticated scans, constraints and energy surface analysis.	Very expensive, carries severe licensing restrictions.
Excellent solvation and reaction field support.	Does not scale to more than 4 (CISD) to 32 (DFT) cores, even on shared memory systems.
Very user-friendly, clear and forgiving input syntax.	A real pain to install (for Linux and OSX)

Gaussian is not the most CPU-efficient package in existence, but it takes much less time to set up a Gaussian job than a job for any other expert level package.

N.B.: CPU time is \$0.01-0.02 an hour, brain time is \$25+ an hour.

What Sets Gaussian 16 Apart from Other Software?

- ◆ Gaussian 16 produces accurate, reliable and complete models without cutting corners.
- ◆ A wide variety of methods makes Gaussian 16 applicable to a broad range of chemical conditions, problem sizes and compounds.
- ◆ Gaussian 16 provides state-of-the-art performance in single CPU, multiprocessor and multicore, cluster/network and GPU computing environments.
- ◆ Setting up calculations is simple and straightforward, and even complex techniques are fully automated. The flexible, easy-to-use options give you complete control over calculation details when needed.
- ◆ Calculation results are presented in natural and intuitive graphical form by GaussView 6.

Gaussian & GaussView

- Useful Links:
- <https://gaussian.com/techsupport/>

White Papers and Technical Notes

- ◆ Creating UV/Visible Plots from the Results of Excited States Calculations
- ◆ Modeling Antiferromagnetic Coupling in Gaussian
- ◆ Transition State Optimizations with Opt=QST2
- ◆ Using Gaussian to Teach Physical Chemistry
- ◆ Comparing NMR Methods in ChemDraw and Gaussian
- ◆ Vibrational Analysis in Gaussian
- ◆ Thermochemistry in Gaussian
- ◆ Visualizing Results when Gaussian and GaussView are on Different Machines
- ◆ Studying Chirality with Vibrational Circular Dichroism
- ◆ Investigating the Reactivity and Spectra of Large Molecules with ONIOM

Historical Documents

- ◆ G09 Citation
- ◆ Versions of Gaussian
- ◆ G09 Keyword pages (gzipped tar archive)
- ◆ GaussView 5 Reference (zipped)

Support

This page contains links to information about obtaining technical support from help@gaussian.com as well as many other resources.

- ◆ Contacting Gaussian, Inc. Technical Support
- ◆ Gaussian Technical support in Mandarin (详细说明点这里)
- ◆ The Gaussian Maintenance Program
- ◆ PC/Mac Product Registration

Gaussian 16 Documentation

- ◆ Release Notes
- ◆ Keyword List
- ◆ G16 Users Reference
- ◆ G16 IOps Reference

GaussView 6 Help Documentation

- ◆ GaussView 6 Help

Installation Instructions

- ◆ Gaussian 16 source code
- ◆ Gaussian 16 UNIX binary
- ◆ Gaussian 16 Mac OS X binary
- ◆ Gaussian 16 for Windows
- ◆ GaussView 6 for UNIX
- ◆ GaussView 6 for Mac OS X
- ◆ GaussView 6 for Windows
- ◆ GMMX 3 for Windows

Linda Documentation

Linda Manual (gzipped)

Gaussian Input Explained

```
%Mem=16GB
%NProcShared=8
#p opt=tight rb3lyp/cc-pVDZ scf=(tight,dsymm)
   integral=(grid=ultrafine)
```

Flavin anion geometry optimization

```
-1 1
C      0.00000000    0.00000000    0.00000000
C      0.00000000    0.00000000    1.41658511
C      1.23106864    0.00000000    2.09037416
C      2.43468421    0.00000000    1.39394037
```

<https://gaussian.com/keywords/>

<https://gaussian.com/dft/>

<https://gaussian.com/basissets/>

<http://gaussian.com/scrf/>

<http://gaussian.com/capabilities/>

-- defines the RAM memory

-- define the number of cores to use

--defines the calculation:

- geometry optimization with tight thresholds (opt=tight)
- Spin Restricted DFT calculation using B3LYP exchange-correlation functional (rb3lyp)
- Use of correlation consistent pVDZ basis sets (cc-pVDZ)
- Requires tight convergence thresholds and symmetrize the density matrix at every SCF iteration (scf=(tight,dsymm))
- Requests ultrafine integration grid for the integrals (integral=(grid=ultrafine))
- total charge =-1 (anion) and spin Multiplicity=1 (i.e. S=0, singlet)

The coordinates of all atoms follow.

Gaussian Output --not-- Explained

The output of Gaussian (i.e. the log file) is quite clumsy, and not intuitive and not user-friendly. Therefore, it is recommended to use GaussView to analyze the results. Gaussian checkfile (*.chk) contains all raw information needed for restart and further processing. It is binary, but it can be made in text using “**formchk**” utility. After the calculation – open the *.chk (or *.fchk) using the GaussView and Click on the Results tab.

```
SCF Done: E(RB3LYP) = -2692.65912953 A.U. after 9 cycles
      NFock= 9 Conv=0.66D-08 -V/T= 2.0012
KE= 2.689322928391D+03 PE=-6.926417431442D+03 EE= 1.295554650233D+03
Leave Link 502 at Thu Aug 19 15:24:52 2021, MaxMem= 104857600 cpu: 65.5 elap: 5.7
(Enter /appl/centos6.3/gnu/apps/gaussian/tiger2/g16/l801.exe)
DoSCS=F DFT=T Scale2(SS,OS)= 1.000000 1.000000
Range of M.O.s used for correlation: 1 99
NBasis= 99 NAE= 30 NBE= 30 NFC= 0 NFV= 0
NROrb= 99 NOA= 30 NOB= 30 NVA= 69 NVB= 69
Leave Link 801 at Thu Aug 19 15:24:53 2021, MaxMem= 104857600 cpu: 0.9 elap: 0.1
(Enter /appl/centos6.3/gnu/apps/gaussian/tiger2/g16/l1101.exe)
Using compressed storage, NAtomX= 11.
Will process 12 centers per pass.
R6Disp: Adding Grimme-D3(BJ) dispersion energy 1st derivatives to the gradient.
Leave Link 1101 at Thu Aug 19 15:24:53 2021, MaxMem= 104857600 cpu: 2.2 elap: 0.2
(Enter /appl/centos6.3/gnu/apps/gaussian/tiger2/g16/l1102.exe)
Symmetrizing basis deriv contribution to polar:
IMax=3 JMax=2 DiffMx= 0.00D+00
Leave Link 1102 at Thu Aug 19 15:24:53 2021, MaxMem= 104857600 cpu: 0.7 elap: 0.1
(Enter /appl/centos6.3/gnu/apps/gaussian/tiger2/g16/l1110.exe)
Forming Gx(P) for the SCF density, NAtomX= 11.
Integral derivatives from FoFJK, PRISM(SPDF).
Do as many integral derivatives as possible in FoFJK.
G2DrvN: MDV= 104856672.
G2DrvN: will do 12 centers at a time, making 1 passes.
Calling FoFCou, ICntrl= 3107 FMM=F IICent= 0 AccDes= 0.00D+00.
FoFJK: IHMeth= 1 ICntrl= 3107 DoSepK=F KAlg= 0 IICent= 0 FoldK=F
```

ORCA



PRO	CONTRA
The best package in existence for spectroscopic properties.	No symmetry factorization.
Simple and forgiving input syntax.	Closed-source package.
Free (not open-source).	No native graphical viewer

```
! UHF HF
%basis basis SV
    newgto Cl "DUNNING-DZP" end
    end
* xyz -2 2
Cu 0 0 0 newgto "TZVPP" end
Cl 2.25 0 0
Cl -2.25 0 0
Cl 0 2.25 0
Cl 0 -2.25 0
*
```

ORCA is the package of choice for NMR and EPR properties, particularly the “difficult” ones, such as ZFS, exchange interaction, *etc.*

```
! MP2 RHF TZVP TightSCF
%mp2      MaxCore 100
          end
%paras    rCO = 1.20
          ACOH = 120
          rCH = 1.08
          end
* int 0 1
C(1) 0 0 0 0.00 0.0 0.00
O(2) 1 0 0 {rCO} 0.0 0.00
H(3) 1 2 0 {rCH} {ACOH} 0.00
H(3) 1 2 3 {rCH} {ACOH} 180.00
*
```

(OPEN)MOLCAS

<https://pubs.acs.org/doi/10.1021/acs.jctc.9b00532>

<https://gitlab.com/Molcas/OpenMolcas>

<https://www.molcas.org/documentation/manual/>

<https://www.molcas.org/tutorials.html>

<https://cobalt.itc.univie.ac.at/molcasforum/>



PRO	CONTRA
The best package in existence for multiconfigurational methods CASSCF/CASPT2/RASSI	not well parallelized
Simple and forgiving input syntax.	No native graphical viewer
Free and open source	

MOLCAS is the package of choice for multiconfigurational calculations: CASSCF, RASSCF, CASPT2, SO-RASSI, ANISO, etc. Transition metals, lanthanides or radicals containing open-shells → MOLCAS can handle

VASP - solid state ab initio package

<https://www.vasp.at>

<https://www.vasp.at/wiki/index.php>

Many tutorials online:

https://www.youtube.com/results?search_query=VASP

Utilities:

<https://jp-minerals.org/vesta/en/>

<https://pymatgen.org>

<https://vaspkit.com>

<https://theory.cm.utexas.edu/vtsttools/scripts.html>



VASP can perform a wide range of calculations for solid state materials at DFT, DFPT, TDDFT, RPA, and GW level of theories: optimise structure, optimise band gaps of insulators, DOS, PDOS, dielectric properties, optical response, phonon calculations, etc.

PRO	CONTRA
One of the best packages for solid-state calculations	expensive
Simple input and very accurate pseudopotentials	No native graphical viewer
well parallelized and fast development	proprietary. Source code available only for licensed groups

Other scientific codes: GAMESS

Notable Pro

Excellent configuration interaction module.

Excellent detailed documentation and very clear input syntax.

Free (not open-source in the GPL sense).

Notable Contra

Three distinct code branches (UK, US, PC) with incompatible syntax.

Does not generally scale to more than 32 cores, even on shared memory systems.

A real pain to install on large-scale systems.

GAMESS offers very fine control over each program option, but requires detailed understanding of the algorithms and hardware.

The particular strength of GAMESS is in the “traditional” (*i.e.* HF, MP_n, CI, MCSCF, *etc.*) quantum chemistry methods.

The logo for GAMESS, featuring the word "GAMESS" in a stylized, bold, serif font with a 3D effect.

```
$CONTRL SCFTYP=RHF RUNTYP=OPTIMIZE COORD=UNIQUE
        ICHARG=0 MULT=1 MAXIT=100 UNITS=ANGS $END
$SYSTEM MWORDS=50 TIMLIM=60000 $END
$BASIS  GBASIS=N311 NGAUSS=6 NPFUNC=1 NDFUNC=1
        DIFFS=.TRUE. DIFFSP=.TRUE. $END
$GUESS  GUESS=HUCKEL PURIFY=.TRUE. $END
$STATPT NSTEP=150 HESS=READ DXMAX=0.05 $END
$SCF    CONV=1E-6 DIRSCF=.TRUE. FDIFF=.TRUE.
        SHIFT=.TRUE. $END
$DFT    METHOD=GRID DFTTYP=B3LYP SWITCH=1E-4
        NRAD=96 NTHE=36 NPHI=72 $END
$DATA
Geometry location run
Cn 2
C 6.0 0.0000000000 0.0000000000 -0.8917392866
C 6.0 1.2134878362 -0.0218104556 -0.2012005677
```

Other scientific codes: ADF

Notable Pro

Slater type basis sets.

Excellent treatment of heavy and transition metals.

Excellent relativistic DFT implementation.

Notable Contra

Fails to converge with many systems containing actinides and lanthanides.

Does not support non-DFT calculations.

Numerous, voluminous and poorly documented input / output files.

```
TITLE Geometry optimization of cyclohexane
XC
  LDA Apply VWN
  GGA Apply BLYP
END
MAXMEMORYUSAGE 30
SCF
  ITERATIONS 250
  CONVERGE 1E-7 1E-7
  MIXING 0.2
  DIIS N=10 OK=0.5 cyc=5 CX=5.0 BFAC=0
END
INTEGRATION 6.0 6.0
atoms
C -4.8948 1.9727 -0.2565
C -3.6335 1.2447 0.2567
```

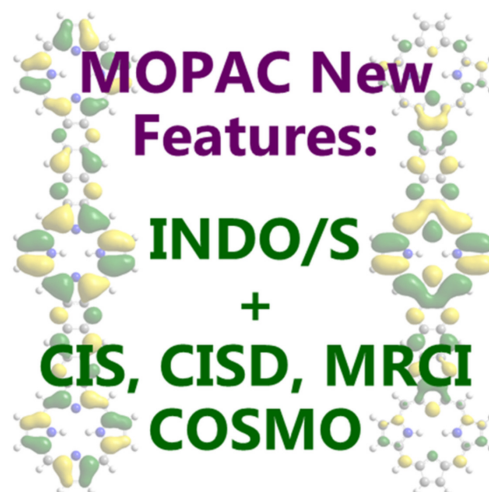


Other scientific codes: MOPAC

<http://openmopac.net>

Also included in many other packages:

ADF, Sparkle, Spartan, ChemDraw, etc.



Notable Pro

Can handle huge systems quantum mechanically.

Some versions are free and open-source.

Includes the latest PM6 semi-empirical method.

Notable Contra

Requires a third-party GUI .

Uncertain licensing status.

Only includes semi-empirical methods.

```
singlet vectors local bonds mullik symmetry
formaldehyde MNDO optimization with symmetry
with local orbital analysis
C
O  1.200000  1
H  1.080000  1  120.000000  1
H  1.080000  0  120.000000  0  180.000000  0
O  0.000000  0  0.000000  0  0.000000  0
  3,  1,  4,
  3,  2,  4,
```

```
1  0  0
1  2  0
1  2  3
0  0  0
```

MOPAC is one of the oldest semi-empirical simulation packages in existence. Many other codes (notably GAMESS) have an interface to MOPAC in their semi-empirical methods modules.

Other scientific codes

<i>Package</i>	<i>Notes</i>
Dalton	Very sophisticated configuration interaction and coupled cluster module.
Turbomole	Probably the fastest and the leanest package in existence.
NWChem	Extremely well parallelized (thousands of cores) – much recommended.
Espresso	DFT code focused on materials. Well parallelized.
Siesta	Expert-friendly DFT package.
Jaguar	Well adapted for transition metal systems.
ONETEP	Linear-scaling DFT code with an emphasis on large systems.
CASTEP	Plane wave DFT code focused on materials. Well parallelized.

On HPC there are many packages installed. You may load them using
module load <module name>

Installation scripts of OPENMOLCAS, VASP and ORCA for HPC and NSCC clusters are provided in LUMINUS.

```
-- download the script from Luminus and transfer to the respective cluster
-- make it executable ( chmod +x <script name> ) and
-- clean Windows line endings ( dos2unix <script name> )
-- run it as usual: ( ./script )
```

Some user-level packages: Material Studio

Notable Pro

Designed specifically for materials research.

Reads and visualizes log files for a large number of packages.

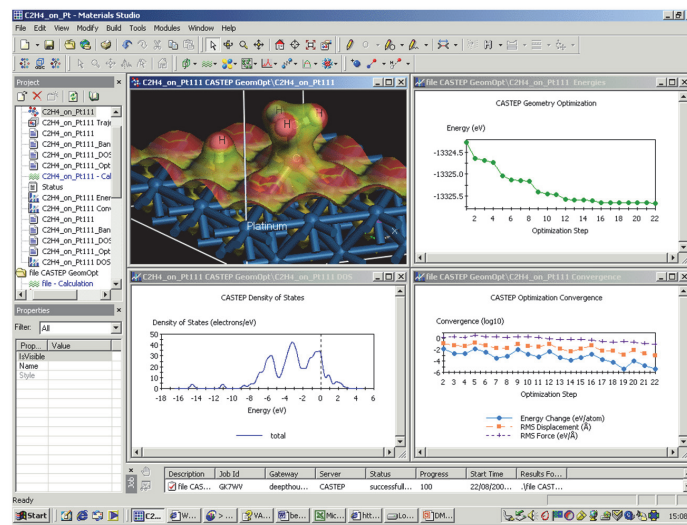
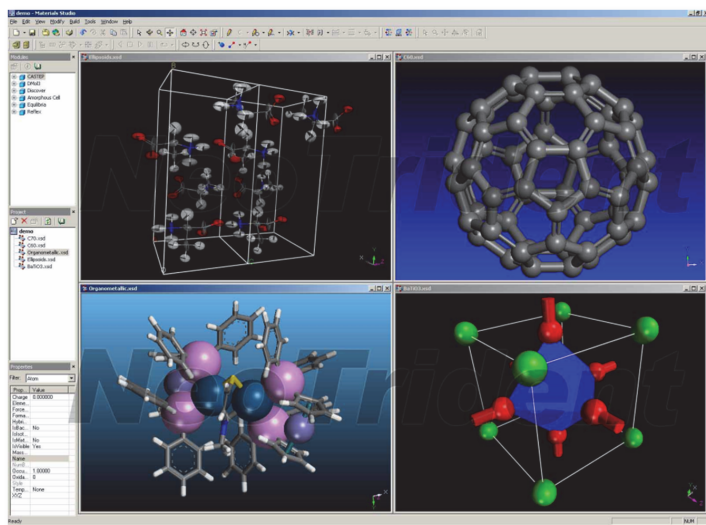
Can output input files and manage jobs for Gaussian, CASTEP, *etc.*

Notable Contra

Inconvenient user interface.

Very expensive with very restrictive licensing.

Licensing system is buggy.



You may use it in NUS. Installation for Windows is available on NUS-IT webpage. Also, you may also run CASTEP jobs on HPC.

Other scientific codes

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-- run it as usual: ( ./script )
```

ArgusLab

Notable Pro

Freely licensed.

Beautiful graphics and 3D export options.

Has an interface to Gaussian and HyperChem.

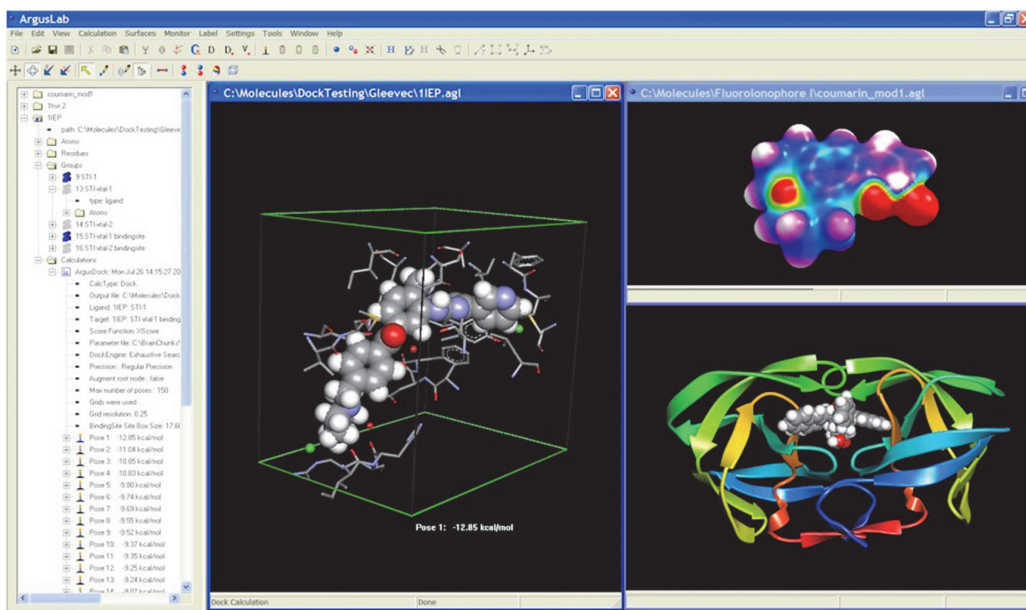
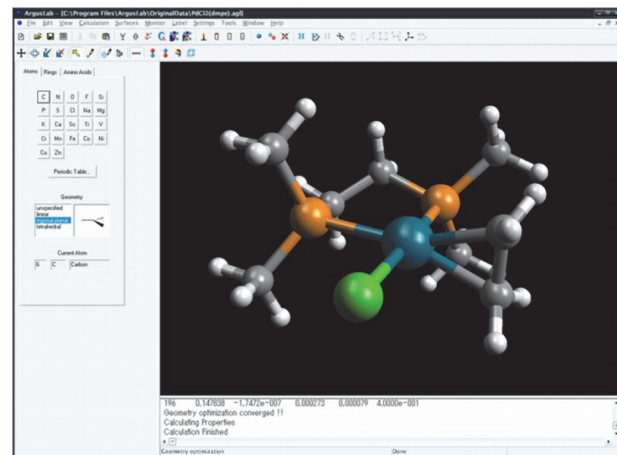
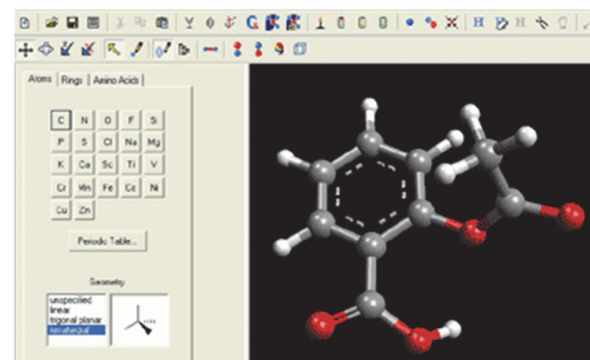
Notable Contra

Very basic user interface.

Development stopped in 2005.

Only the most basic methods included.

ArgusLab

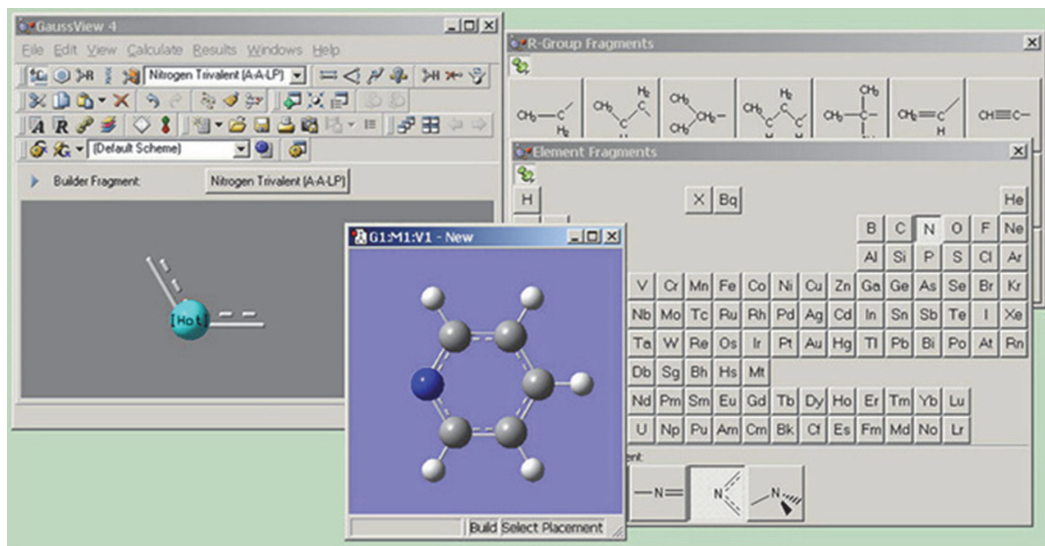


GaussView 6

PRO	CONTRA
A truly outstanding GUI – very powerful and convenient.	So good it's addictive.
Can set up Gaussian jobs and interpret Gaussian output.	Quite expensive.
Supports all major molecule specification formats.	Can only export 2D images.

<http://gaussian.com/gv611rn/>

Please follow this link to learn more about the capabilities of the GaussView software



GaussView is a must-have for most types of research involving QC calculations.

Running GaussView 6 remotely on HPC

Students can access Gaussview remotely with X11 forwarding.
For macOS users: install XQuartz (free, <https://www.xquartz.org/>),
then from Terminal:

```
ssh -X <username>@vanda.nus.edu.sg  
source /app1/ebenv  
module load Gaussian/16.C.02-AVX2  
/app1/common/Gaussian/Rev-C.02/gv/gview.sh
```

This opens the full GaussView GUI, displayed locally on their Mac while running on the cluster — so students can still complete their GUI-based coursework without needing a local install.

Please note that wall time on the vanda cluster is limited by allocated credits, so extended GaussView usage may be affected.

Normally, everyone has some CPU hours by default on **vanda**. if not – please request initialization CPU time.

MOLDEN

Notable Pro

Supports a large number of simulation packages.

Licensed freely.

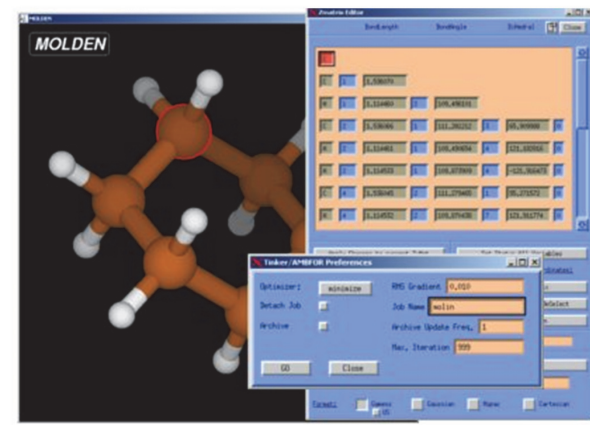
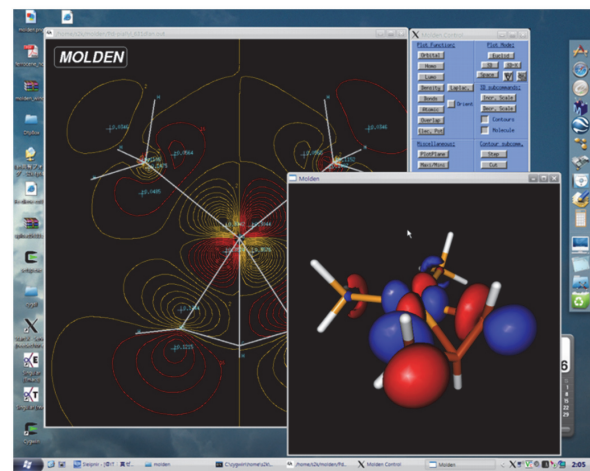
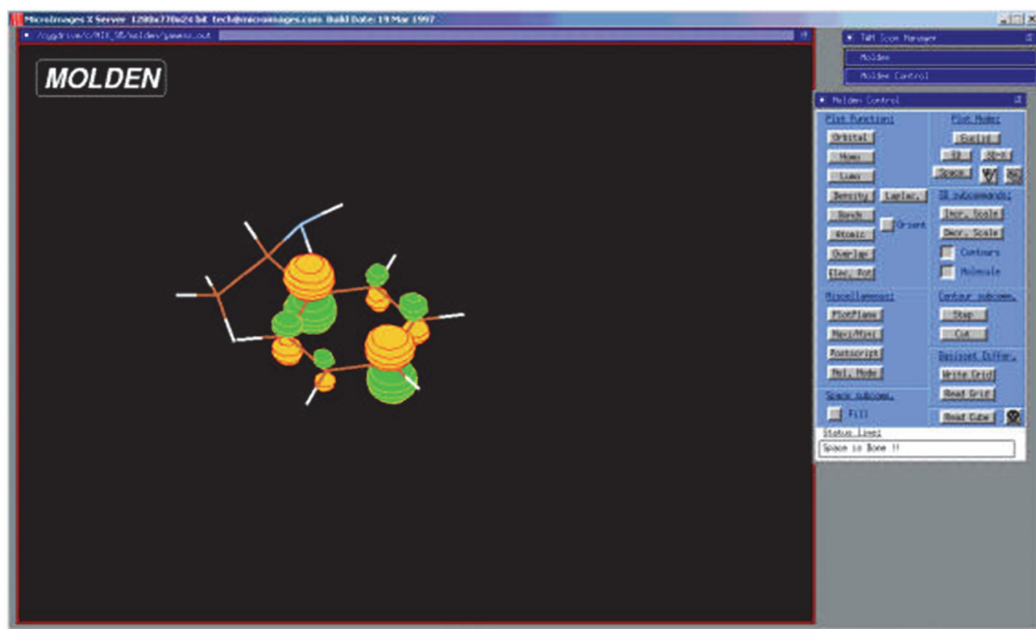
Outputs 3D graphics.

Notable Contra

Difficult to run on Windows.

Structure editing is difficult.

Very old school user interface.



JMOL

Notable Pro

Free and open-source.

Outputs 3D graphics.

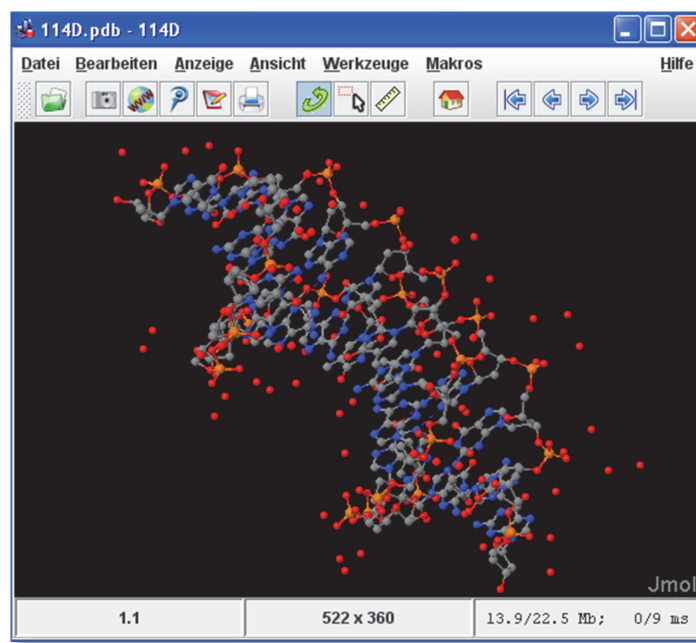
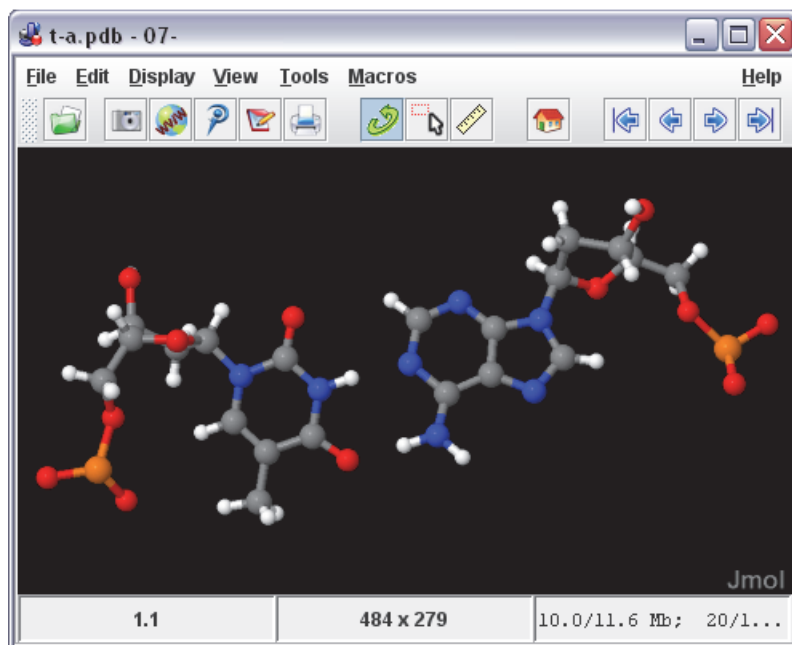
Supports a large number of simulation packages.

Notable Contra

Runs under Java with all of the consequent problems.

Very basic user interface.

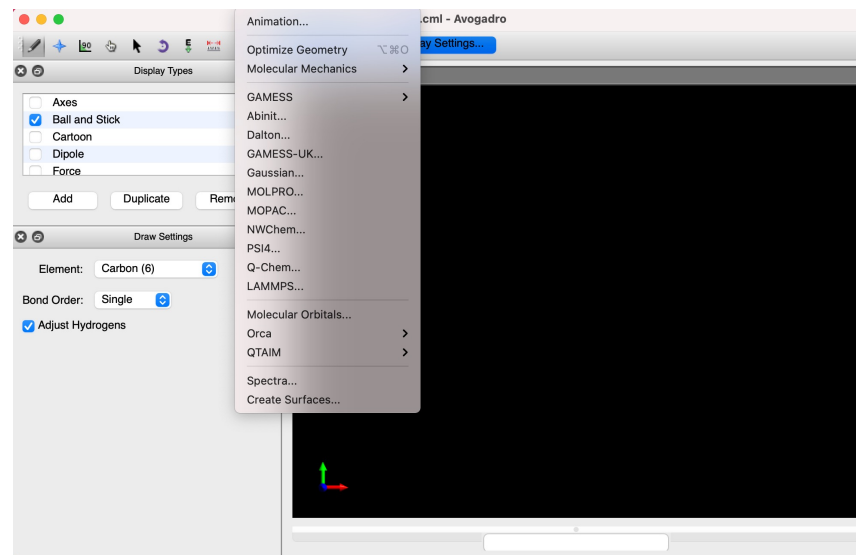
No structure editing capabilities.



Avogadro

<https://avogadro.cc>

PRO	CONTRA
Free and open source	Basic structure editing
Outputs 3D graphics	Basic User interface
Supports a large number of packages	



We will use AVOGADRO to inspect the results of ORCA calculations, e.g. vibrational frequencies.

ORCA print level must be set to **“large”**. Avogadro know where to find the required info from the ORCA output.

CHEMCRAFT

<http://www.chemcraftprog.com>

Notable Pro

Excellent structure editing capabilities.

Dedicated support for GAMESS, Gaussian and ADF.

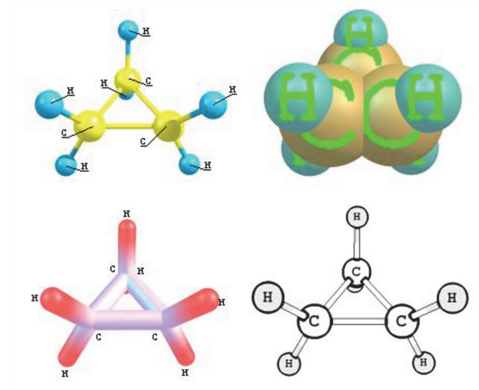
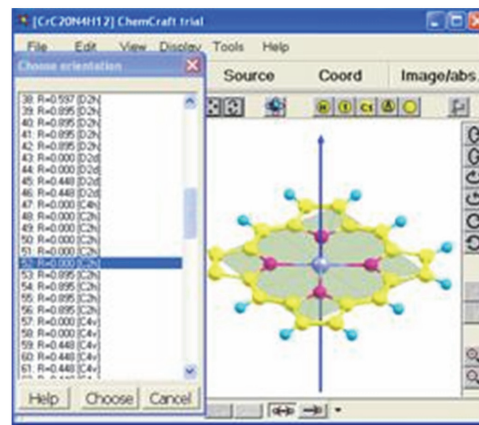
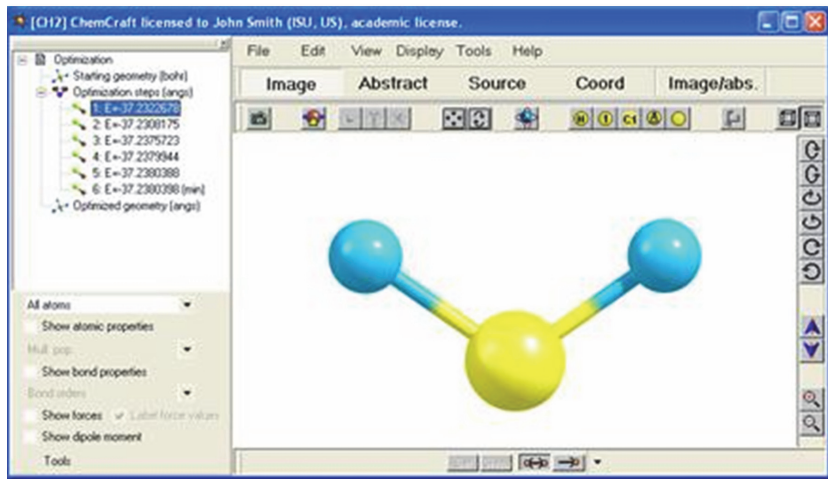
Outputs animations and calculation inputs.

Notable Contra

Source code not available.

Commercial package (\$100-\$200 depending on the license)

Old school user interface.



Trial Version of *CHEMCRAFT* is for 6 months (180 days)!
It supports ORCA.

VMD

<http://www.ks.uiuc.edu/Research/vmd/>

Notable Pro

Very flexible large-scale structure visualizer.

Outputs 3D graphics.

Supports all major molecule specification formats.

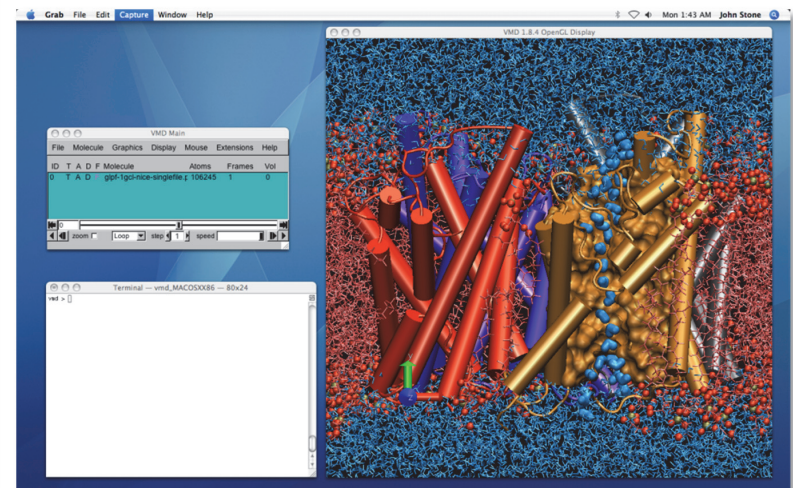
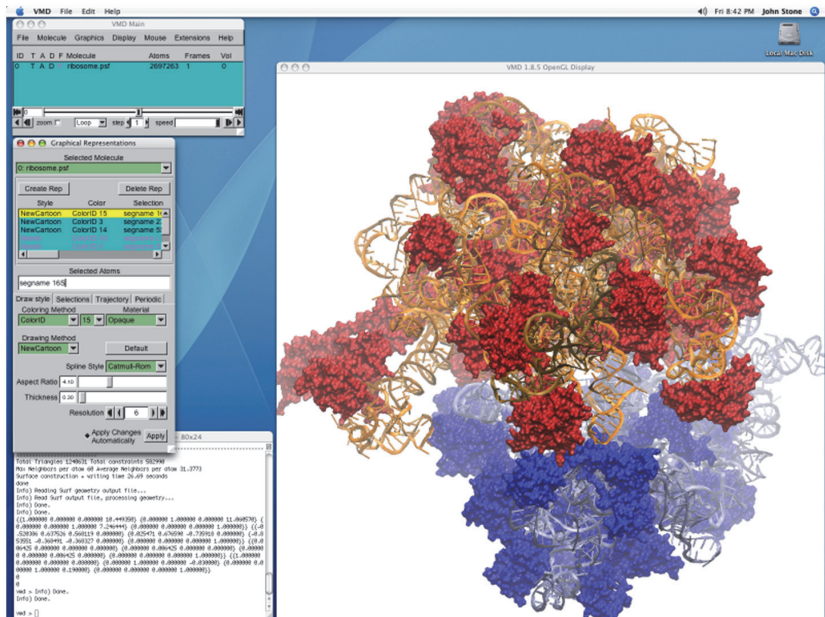
Notable Contra

Specifically targets large biomolecules.

Lacks structure editing capabilities.

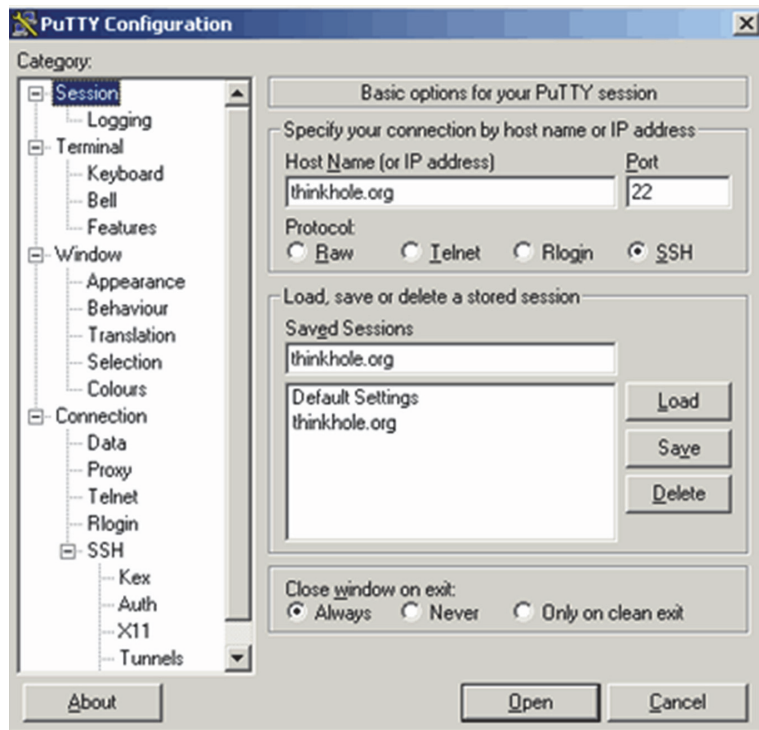
Very old school user interface.

VMD
Visual Molecular Dynamics



SSH login and file transfer

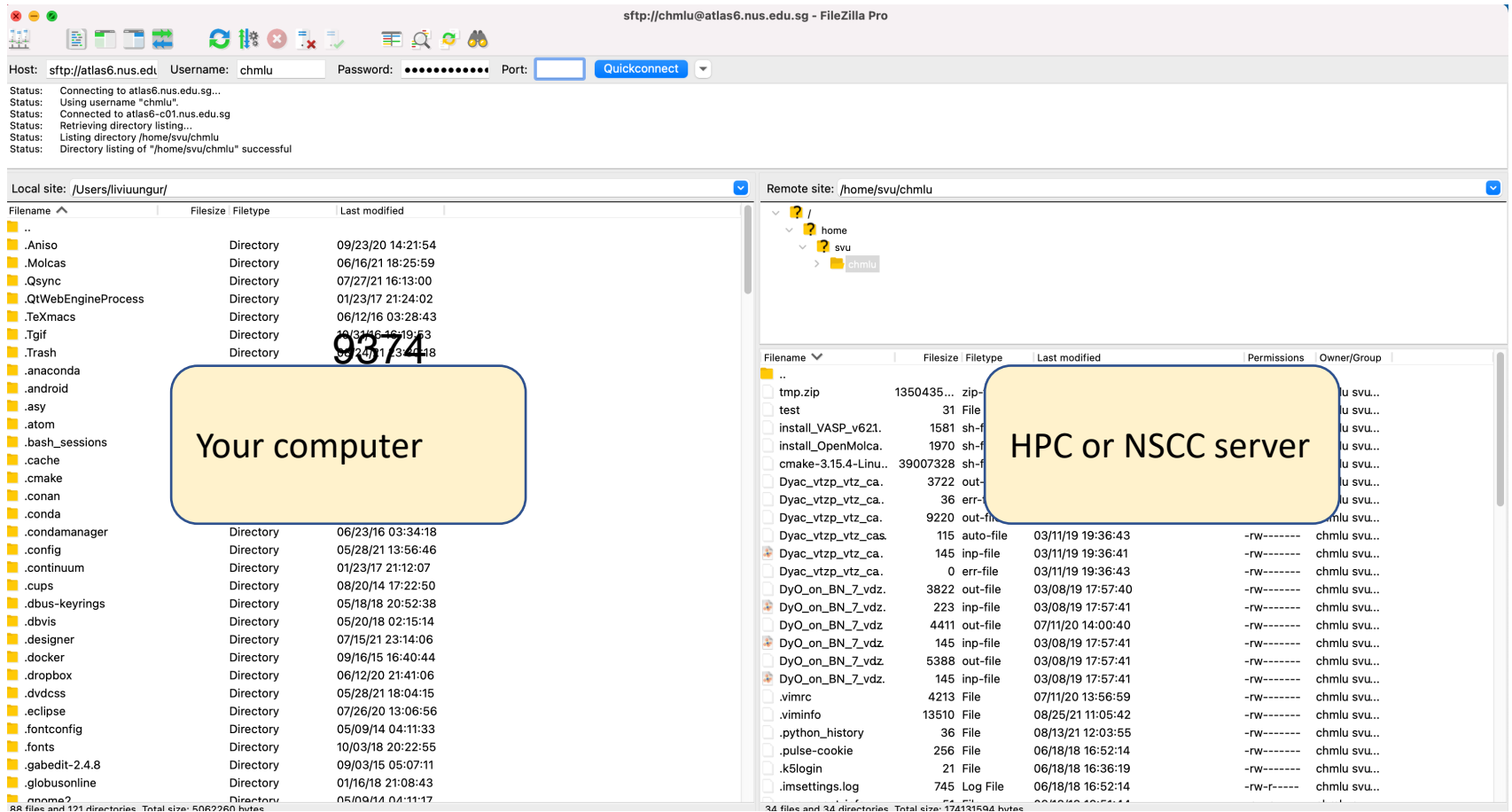
Putty + WinSCP or FileZilla



Putty is one of the best *ssh client* for Windows in existence. If you are using Windows, it would be wise to familiarize yourself with its various options.

There are several alternatives:
PowerShell
XShell
etc.

SSH file transfer: FileZilla or WinSCP



Drag-and-drop easily files between your computer and servers. You may even edit files from the server → install **Notepad++**

Variational Principle

(one of the most important theorems used in Computational Chemistry)

Variational Principle

For any operator, when we use its own ground eigenstate – we get the lowest possible eigenvalue.

$$E_{exact} = \frac{\langle \psi_{exact} | \hat{H} | \psi_{exact} \rangle}{\langle \psi_{exact} | \psi_{exact} \rangle}$$

When we use an approximate function (i.e. not its eigenstate) to compute the expectation value of the operator – the result will always be higher.

$$E_{trial} = \frac{\langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle}$$

The above theorem states that: $E_{trial} \geq E_{exact}$

Proof:

Expand the trial function as a linear combination of exact eigenstates of the Hamiltonian (= decompose ψ_{trial} in the basis set of the Hamiltonian):

$$\psi_{trial} = \sum_{n=1} c_n \varphi_n, \quad \text{where} \quad \hat{H} \varphi_n = \varepsilon_n \varphi_n$$

$$\begin{aligned} E_{trial} &= \langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle = \sum_{n=1} \sum_{m=1} c_m^* c_n \langle \varphi_n | \hat{H} | \varphi_m \rangle = \sum_{n=1} \sum_{m=1} c_m^* c_n \langle \varphi_m | \hat{H} \varphi_n \rangle = \\ &= \sum_{n=1} \sum_{m=1} c_m^* c_n \varepsilon_n \langle \varphi_m | \varphi_n \rangle = \sum_{n=1} \sum_{m=1} c_m^* c_n \varepsilon_n \delta_{mn} = \sum_{n=1} c_n^* c_n \varepsilon_n \end{aligned}$$

On the other hand:

$$E_{exact} = \langle \psi_{exact} | \hat{H} | \psi_{exact} \rangle = \langle \varphi_1 | \hat{H} | \varphi_1 \rangle = \varepsilon_1$$

Variational Principle

On the other hand:

$$E_{exact} = \langle \psi_{exact} | \hat{H} | \psi_{exact} \rangle = \langle \varphi_1 | \hat{H} | \varphi_1 \rangle = \varepsilon_1$$

The difference:

$$E_{trial} - E_{exact} = \left(\sum_{n=1} c_n^* c_n \varepsilon_n \right) - \varepsilon_1 = \left(\sum_{n=1} c_n^* c_n (\varepsilon_n - \varepsilon_1) \right) \geq 0$$

The inequality follows from $c_n^* c_n \geq 0$ and $\varepsilon_n \geq \varepsilon_1$.

Variational Method in practice

It is quite common that that we do not have the exact wave function, but we are able to build an approximate function. Usually, approximate wave functions have some parameters which can be adjusted during the calculation.

Step 1: Build the trial wave function

$$\psi_{trial} = \sum_{n=1} c_n \varphi_n$$

Step 2: Calculate the trial value of energy using the formula:

$$E_{trial} = \frac{\langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle}$$

Step 3: Set up ALL first derivatives of the energy wrt variational coefficients equal to zero.

$$\frac{\partial E_{trial}}{\partial c_i} = 0$$

Solve the obtained system of equations with respect to the variational parameters $\{c_i\}$.

Step 4: Use the obtained set of $\{c_i\}$ to calculate the final energy E_{trial_best}

The set of parameters $\{c_i\}$ which satisfy the condition at step 3 are called the **solution**. They represent the BEST variational parameters for the ψ_{trial} , while the obtained energy is the lowest possible.

Further improvement can only be done if:

- a) the number of variational parameters is increased (expand the basis set of functions)
- b) change the wave function representation, which also includes an increase in the number of variational parameters to be optimized.

Non-linear variation: H atom

Ground state of H atom is:

$$\psi_{1s} = \frac{1}{\sqrt{\pi}} e^{-r}, \quad E_{1s} = -\frac{1}{2} \text{ (a. u.)}$$

Suppose we did not know this function. We could assume that a good function describing H atom was:

$$\psi_{trial} = \sqrt{\frac{\alpha^3}{\pi}} e^{-\alpha r}$$

What is the value of parameter α which gives the lowest energy?

Non-linear variation: H atom

Solution:

1) Check if the ψ_{trial} is normalized.

$$\begin{aligned}\int \psi_{trial}^* \psi_{trial} d\tau &= \frac{\alpha^3}{\pi} \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi \int r^2 e^{-2\alpha r} dr = \\ &= 4\pi \frac{\alpha^3}{\pi} \frac{2}{(2\alpha)^3} = 1\end{aligned}$$

This means that the trial wave function is normalized.

2) Calculate the energy using this trial function:

$$E = \int \psi_{trial}^* \hat{H} \psi_{trial} d\tau$$

The effect of the $\hat{H}\psi_{trial}$ can be computed separately:

Non-linear variation: H atom

The effect of the $\hat{H}\psi_{trial}$ can be computed separately:

$$\hat{H}\psi_{trial} = \left(-\frac{1}{2} \frac{1}{r^2} \frac{d}{dr} r^2 \frac{d}{dr} - \frac{1}{r} \right) \sqrt{\frac{\alpha^3}{\pi}} e^{-\alpha r} = \dots = \left[\frac{\alpha - 1}{r} - \frac{\alpha^2}{2} \right] \sqrt{\frac{\alpha^3}{\pi}} e^{-\alpha r}$$

Include this result into the Energy expression:

$$\begin{aligned} E &= \int \psi_{trial}^* \hat{H} \psi_{trial} d\tau = 4\pi \times \frac{\alpha^3}{\pi} \int \left[\frac{\alpha - 1}{r} - \frac{\alpha^2}{2} \right] r^2 e^{-2\alpha r} dr \\ &= 4\alpha^3(\alpha - 1) \int r e^{-2\alpha r} dr - 2\alpha^5 \int r^2 e^{-2\alpha r} dr = \dots = \frac{\alpha^2}{2} - \alpha \end{aligned}$$

Non-linear variation: H atom

3) Calculate the derivative of energy with respect to parameter α

$$\frac{dE}{d\alpha} = \alpha - 1$$

Setting the $\frac{dE}{d\alpha} = 0$, we find $\alpha = 1$

$$\psi_{trial} = \sqrt{\frac{1}{\pi}} e^{-r} = \psi_{1s}$$

In this case we were lucky that the form of the trial wave function can become the true wave function.

Assume another case when ψ_{trial} can not become the true ψ_{1s} .

$$\psi_{trial} = \sqrt{\frac{\alpha^5}{3\pi}} r e^{-\alpha r}$$

Exercise: Prove that ψ_{trial} is normalized.

Non-linear variation: H atom

Assume another case when ψ_{trial} can not become the true ψ_{1s} .

$$\psi_{trial} = \sqrt{\frac{\alpha^5}{3\pi}} r e^{-\alpha r}$$

Exercise: Prove that ψ_{trial} is normalized.

Proceeding as before:

$$\hat{H}\psi_{trial} = \left(-\frac{1}{2} \frac{1}{r^2} \frac{d}{dr} r^2 \frac{d}{dr} - \frac{1}{r} \right) \sqrt{\frac{\alpha^5}{3\pi}} r e^{-\alpha r} = \dots = \left[-\frac{1}{r^2} + \frac{2\alpha - 1}{r} - \frac{\alpha^2}{2} \right] \sqrt{\frac{\alpha^5}{3\pi}} r e^{-\alpha r}$$

Computing the energy we get:

Non-linear variation: H atom

Computing the energy we get:

$$E = \int \psi_{trial}^* \hat{H} \psi_{trial} d\tau = 4\pi \frac{\alpha^5}{3\pi} \int \left[-\frac{1}{r^2} + \frac{2\alpha - 1}{r} - \frac{\alpha^2}{2} \right] r^4 e^{-2\alpha r} dr = \dots = \frac{4}{3} \left[\frac{\alpha^2}{8} - \frac{3\alpha}{8} \right]$$

The first derivative wrt α is:

$$\frac{dE}{d\alpha} = 0 = \frac{4}{3} \left[\frac{\alpha}{4} - \frac{3}{8} \right]$$

Which leads to $\alpha = \frac{3}{2}$

The improved wave function is therefore:

$$\psi'_{trial} = \sqrt{\frac{\alpha^5}{3\pi}} r e^{-\alpha r} = \sqrt{\frac{243}{96\pi}} r e^{-\frac{3}{2}r}$$

Non-linear variation: H atom

Computing the energy using the improved wave function, we get:

$$E = \frac{4}{3} \left[\frac{\alpha^2}{8} - \frac{3\alpha}{8} \right] = \frac{4}{3} \left[\frac{(3/2)^2}{8} - \frac{3(3/2)}{8} \right] = -\frac{3}{8} \text{ (a.u.)}$$

This energy is above the lowest eigenvalue ($-\frac{1}{2}$ a.u.) but is well below the second eigenvalue ($-\frac{1}{8}$ a.u.).

The energy is close to the true energy of $1s$ state.

What is the weight of the true $1s$ contribution into the obtained ψ'_{trial} ?

General wave function: linear combination of basis functions

Our ψ'_{trial} can be understood as a superposition of many states. We can expand it in a series of orthogonal eigenstates of the Hamiltonian.

$$\psi'_{trial} = c_{1s}\psi_{1s} + c_{2s}\psi_{2s} + c_{3s}\psi_{3s} + c_{2p_x}\psi_{2p_x} + c_{2p_y}\psi_{2p_y} + c_{2p_z}\psi_{2p_z} + \dots$$

Projection operation:

We find the coefficients of individual ψ_k by multiplying ψ'_{trial} with the ψ_k^* from the left. As example:

$$\int \psi_{1s}^* \psi'_{trial} d\tau = \int \psi_{1s}^* c_{1s} \psi_{1s} d\tau + \int \psi_{1s}^* c_{2s} \psi_{2s} d\tau + \dots$$

All terms except the one containing the c_{1s} will cancel because the eigenstates of the Hamiltonian are orthogonal to each other. Therefore:

$$c_{1s} = \int \psi_{1s}^* \psi'_{trial} d\tau$$

Projection operation

Compute the c_{1s} explicitly in this case:

$$\begin{aligned} c_{1s} &= \int \left(\frac{1}{\sqrt{\pi}} e^{-r} \right) \left(\sqrt{\frac{243}{96\pi}} r e^{-\frac{3}{2}r} \right) d\tau = 4\pi \frac{1}{\sqrt{\pi}} \sqrt{\frac{243}{96\pi}} \int r^2 (e^{-r}) (r e^{-\frac{3}{2}r}) d\tau = \\ &= \frac{9}{\sqrt{2}} \int r^3 e^{-\frac{5}{2}r} dr = \frac{9}{\sqrt{2}} \frac{6}{(5/2)^4} = \frac{432\sqrt{2}}{625} \approx 0.9975 \end{aligned}$$

As we see, the weight of the $1s$ into our ψ'_{trial} is $c_{1s}^2 = 0.9555$ or 99.55% .

Exercise: calculate the weight of the $2s$ in the above expansion of ψ'_{trial} .

Review of the self-consistent Hartree-Fock method

- Born-Oppenheimer approximation is considered (static nuclei)
- Each electron moves in the average electrostatic field created by all other electrons and nuclei
- Wave function is represented by a *single* Slater Determinant:

$$\psi_{SD} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \varphi_1(1)\sigma(1) & \varphi_2(1)\sigma(1) & \cdots & \varphi_N(1)\sigma(1) \\ \varphi_1(2)\sigma(2) & \varphi_2(2)\sigma(2) & \cdots & \varphi_N(2)\sigma(2) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_1(N)\sigma(N) & \varphi_2(N)\sigma(N) & \cdots & \varphi_N(N)\sigma(N) \end{vmatrix}$$

- $\varphi_1(i), \varphi_2(i), \dots, \varphi_N(i)$ -- molecular orbitals occupied by electron i
- $\sigma(i)$ -- spin function of electron i
- $\varphi_1(i)\sigma(i), \varphi_2(i)\sigma(i)$ -- molecular spin-orbitals
- $\frac{1}{\sqrt{N!}}$ -- normalization constant. There are $N!$ terms in the expansion of a determinant.

$$\int \psi_{SD}^* \psi_{SD} d\tau = \langle \psi_{SD} | \psi_{SD} \rangle = 1$$

Shorthand notation: $\psi_{SD} = |\varphi_1 \varphi_2 \dots \varphi_N\rangle$. A bar on top of $\bar{\varphi}_i$ means spin-orbital $\varphi_i\beta$.

Review of the self-consistent Hartree-Fock method

Example: Wave function for H₂ molecule in a minimum basis set representation.

Basis function on atom H1 = $1s_1$
 Basis function on atom H2 = $1s_2$ } *basis functions on different atoms are not orthogonal to each other, overlap $S_{12} = \int 1s_1^* 1s_2^* d\tau > 0$*

Orthogonal molecular orbitals:

$\varphi_g = \frac{1}{\sqrt{2(1+S_{12})}} (1s_1 + 1s_2)$
 $\varphi_u = \frac{1}{\sqrt{2(1-S_{12})}} (1s_1 - 1s_2)$ } *Orthogonality check: $\int 1\varphi_g^* 1\varphi_u^* d\tau = 0$*

Ground state Slater Determinant:

$$\psi_{H_2} = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_g(1)\alpha(1) & \varphi_g(1)\beta(1) \\ \varphi_g(2)\alpha(2) & \varphi_g(2)\beta(2) \end{vmatrix} \equiv \begin{array}{c} \text{---} \\ \uparrow \downarrow \\ \text{---} \end{array} \begin{array}{l} \varphi_u \\ \varphi_g \end{array}$$

φ_u molecular orbital has no electrons.

The energy of the ground state can be computed as expectation value of the Hamiltonian:

$$E = \langle \psi_{H_2} | \hat{H} | \psi_{H_2} \rangle = \left\langle \psi_{H_2} \left| \sum_i^{N_{elec}} \hat{\mathbf{h}}_i + \sum_{j>i}^{N_{elec}} \hat{\mathbf{g}}_{ij} + V_{nn} \right| \psi_{H_2} \right\rangle =$$

Molecular integrals

$$\psi_{H2} = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_g(1)\alpha(1) & \varphi_g(1)\beta(1) \\ \varphi_g(2)\alpha(2) & \varphi_g(2)\beta(2) \end{vmatrix} = \frac{1}{\sqrt{2}} [\varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) - \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2)]$$

$$E = \langle \psi_{H2} | \hat{H} | \psi_{H2} \rangle = \langle \psi_{H2} | \sum_i^{N_{elec}} \hat{\mathbf{h}}_i + \sum_i^{N_{elec}} \sum_{j>i}^{N_{elec}} \hat{\mathbf{g}}_{ij} + V_{nn} | \psi_{H2} \rangle =$$

$$= \sum_i^{N_{elec}} \langle \psi_{H2} | \hat{\mathbf{h}}_i | \psi_{H2} \rangle + \sum_i^{N_{elec}} \sum_{j>i}^{N_{elec}} \langle \psi_{H2} | \hat{\mathbf{g}}_{ij} | \psi_{H2} \rangle + V_{nn}$$

Evaluate the terms separately. **One-electron** integrals:

$$\langle \psi_{H2} | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \psi_{H2} \rangle$$

$$= \left\langle \frac{1}{\sqrt{2}} [\varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) - \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2)] \right| \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 \left| \frac{1}{\sqrt{2}} [\varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) - \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2)] \right\rangle$$

$$= \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle$$

$$- \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle - \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle =$$

Molecular integrals

$$\begin{aligned}
 &= \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle \\
 &- \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle - \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle = \\
 &= \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle \\
 &+ \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle \\
 &- \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle - \frac{1}{2} \langle \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle \\
 &- \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle - \frac{1}{2} \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(1)\beta(1)\varphi_g(2)\alpha(2) \rangle = \\
 &= \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\beta(2) | \varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\beta(2) \rangle \\
 &+ \frac{1}{2} \langle \varphi_g(1)\beta(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\alpha(2) | \varphi_g(2)\alpha(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1) | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\alpha(2) \rangle -
 \end{aligned}$$

Molecular integrals

$$\begin{aligned}
 &= \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\beta(2) | \varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\beta(2) \rangle \\
 &+ \frac{1}{2} \langle \varphi_g(1)\beta(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\alpha(2) | \varphi_g(2)\alpha(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1) | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\alpha(2) \rangle \\
 &- \frac{1}{2} \langle \varphi_g(1)\beta(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\alpha(2) | \varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1) | \varphi_g(1)\alpha(1) \rangle \langle \varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\beta(2) \rangle \\
 &- \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\beta(2) | \varphi_g(2)\alpha(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \varphi_g(1)\beta(1) \rangle \langle \varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\alpha(2) \rangle = \\
 &= \frac{1}{2} \langle \varphi_g(1)\alpha(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\alpha(1) \rangle + \frac{1}{2} \langle \varphi_g(2)\beta(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\beta(2) \rangle + \frac{1}{2} \langle \varphi_g(1)\beta(1) | \hat{\mathbf{h}}_1 | \varphi_g(1)\beta(1) \rangle + \frac{1}{2} \langle \varphi_g(2)\alpha(2) | \hat{\mathbf{h}}_2 | \varphi_g(2)\alpha(2) \rangle
 \end{aligned}$$

After integration over the spin:

$$= \frac{1}{2} \langle \varphi_g(1) | \hat{\mathbf{h}}_1 | \varphi_g(1) \rangle + \frac{1}{2} \langle \varphi_g(2) | \hat{\mathbf{h}}_2 | \varphi_g(2) \rangle + \frac{1}{2} \langle \varphi_g(1) | \hat{\mathbf{h}}_1 | \varphi_g(1) \rangle + \frac{1}{2} \langle \varphi_g(2) | \hat{\mathbf{h}}_2 | \varphi_g(2) \rangle =$$

Finally:

$$\langle \psi_{H2} | \hat{\mathbf{h}}_1 + \hat{\mathbf{h}}_2 | \psi_{H2} \rangle = \langle \varphi_g(1) | \hat{\mathbf{h}}_1 | \varphi_g(1) \rangle + \langle \varphi_g(2) | \hat{\mathbf{h}}_2 | \varphi_g(2) \rangle = 2 \langle \varphi_g(1) | \hat{\mathbf{h}}_1 | \varphi_g(1) \rangle$$

Since the $\hat{\mathbf{h}}_1 = \hat{\mathbf{h}}_2$ the two terms have the same value.

Molecular integrals

Bielectronic contribution can be evaluated similarly.

$$\begin{aligned}
 \langle \psi_{H_2} | \hat{\mathbf{g}}_{12} | \psi_{H_2} \rangle &= \dots (try) \dots \\
 &= \left[\langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{g}}_{12} | \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) \rangle - \langle \varphi_g(1)\alpha(1)\varphi_g(2)\beta(2) | \hat{\mathbf{g}}_{12} | \varphi_g(2)\alpha(2)\varphi_g(1)\beta(1) \rangle \right] = \\
 &\quad \text{Coulomb Integral} \qquad \qquad \qquad \text{Exchange Integral}
 \end{aligned}$$

After integration over the spin:

$$= \langle \varphi_g(1)\varphi_g(2) | \hat{\mathbf{g}}_{12} | \varphi_g(1)\varphi_g(2) \rangle + 0$$

Finally, the energy of H_2 becomes:

$$E = \langle \psi_{H_2} | \hat{H} | \psi_{H_2} \rangle = 2\langle \varphi_g(1) | \hat{\mathbf{h}}_1 | \varphi_g(1) \rangle + \langle \varphi_g(1)\varphi_g(2) | \hat{\mathbf{g}}_{12} | \varphi_g(1)\varphi_g(2) \rangle$$

Energy of a Slater Determinant

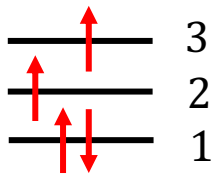
General energy expression for a given Slater Determinant:

$$E = \sum_i^{N_{elec}} h_{ii} + \sum_i^{N_{elec}} \sum_{j>i}^{N_{elec}} (J_{ij} - K_{ij}) + V_{nn}$$

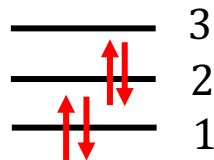
J_{ij} —is non-zero for all pairs of electrons, regardless of their spin

K_{ij} —is non-zero for all pairs of electrons with same spin

Calculate the energy of the following Slater Determinants:



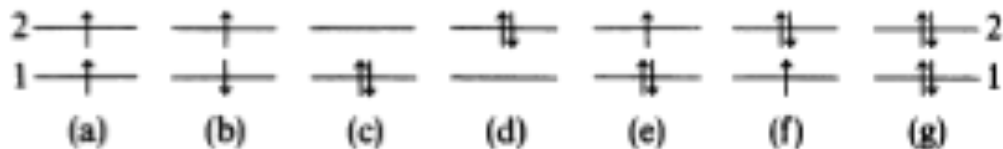
$$E = 2h_{11} + h_{22} + h_{33} + J_{11} + 2J_{12} + 2J_{13} + J_{23} - K_{12} - K_{13} - K_{23}$$



$$E = 2h_{11} + 2h_{22} + J_{11} + J_{22} + 4J_{12} - 2K_{12}$$

Energy of a Slater Determinant

Exercise: Verify the energies of the following Slater determinants by inspection:



- a. $h_{11} + h_{22} + J_{12} - K_{12}$.
 - b. $h_{11} + h_{22} + J_{12}$.
 - c. $2h_{11} + J_{11}$.
 - d. $2h_{22} + J_{22}$.
 - e. $2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$.
 - f. $2h_{22} + h_{11} + J_{22} + 2J_{12} - K_{12}$.
 - g. $2h_{11} + 2h_{22} + J_{11} + J_{22} + 4J_{12} - 2K_{12}$.
-

PS. The two Slater Determinant listed in a) and b) in the above exercise are linked to the so-called Broken-Symmetry HF / DFT: The energy difference between these two Slater Determinants is the “magnetic exchange interaction, K_{12} ”. We will discuss this method in more detail later in the course.

Fock Operator

Fock operator (it defines the effective kinetic energy of an electron and the attraction to all nuclei):

$$\mathbf{F}_i = \mathbf{h}_i + \sum_{j=1}^{N_{elec}} (\mathbf{J}_j - \mathbf{K}_j)$$

HF orbital energies are simply eigenvalues of the $\mathbf{F}_i$ operator:

$$\mathbf{F}_i \varphi_i = \varepsilon_i \varphi_i$$

When HF orbital optimization calculation is finished, φ_i are **canonical** molecular orbitals. Fock operator written in the basis of *canonical* molecular orbitals is diagonal.

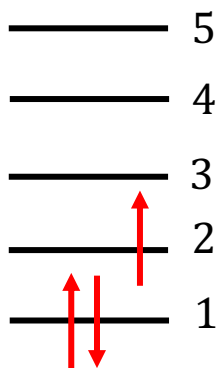
There are many orbital sets possible for the HF wave function, giving the same total energy.

Any orbital set obtained by some **unitary transformation** of the converged molecular orbitals – will also **give the same identical total HF energy**.

Example: **natural orbitals** are obtained by diagonalization of the **one-electron density matrix**

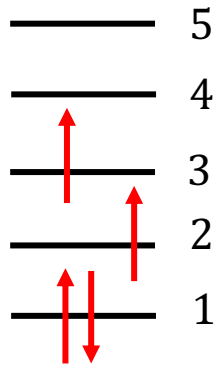
Koopmans Theorem

Let's use the obtained set of molecular orbitals for a system with N electrons to compute the energy of a system with (N+1) and (N-1) electrons. Consider that the orbitals remain exactly the same as in the original calculation.



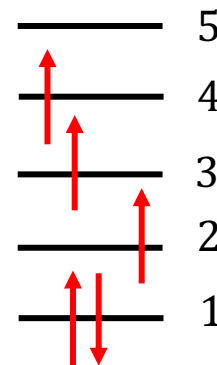
N-1 case

$$E_{N-1} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$$



N case

$$E_N = 2h_{11} + h_{22} + h_{33} + J_{11} + 2J_{12} + 2J_{13} + J_{23} - K_{12} - K_{13} - K_{23}$$



N+1 case

$$E_{N+1} = 2h_{11} + h_{22} + h_{33} + h_{44} + J_{11} + 2J_{12} + 2J_{13} + 2J_{14} + J_{23} + J_{24} + J_{34} - K_{12} - K_{13} - K_{14} - K_{23} - K_{24} - K_{34}$$

Koopmans Theorem

Let's calculate the energy differences between these configurations:

- $E_N - E_{N-1} = (h_{11} + h_{22} + h_{33} + J_{12} + J_{13} + J_{23} - K_{12} - K_{13} - K_{23}) - (h_{11} + h_{22} + J_{12} - K_{12}) = h_{33} + (J_{13} - K_{13}) + (J_{23} - K_{23}) = \mathbf{F}_3$
- $E_{N+1} - E_N = (h_{11} + h_{22} + h_{33} + h_{44} + J_{12} + J_{13} + J_{14} + J_{23} + J_{24} + J_{34} - K_{12} - K_{13} - K_{14} - K_{23} - K_{24} - K_{34}) - (h_{11} + h_{22} + h_{33} + J_{12} + J_{13} + J_{23} - K_{12} - K_{13} - K_{23}) = h_{44} + (J_{14} - K_{14}) + (J_{24} - K_{24}) + (J_{34} - K_{34}) = \mathbf{F}_4$

Observe that adding one electron on the orbital i needs an energy equal to $\mathbf{F}_i =$

Since we are using now *canonical orbitals* (for which orbital energy is well defined), the $\mathbf{F}_i$ is diagonal.

$$\langle \varphi_i | \mathbf{F}_i | \varphi_i \rangle = \varepsilon_i$$

- $E_N - E_{N-1} = \varepsilon_i$

Wave function for B atom

Let's assume that we have the task to calculate the ground state of the free B atom. You have two options:

- 1) You can start your calculation without thinking (any input is rather trivial)
- 2) Do a little exercise and try to understand what is the nature of the wave function. Is the Hartree-Fock wave function adequate? Argument.

Wave function for B atom

Ground electronic configuration of B: $1s^2 2s^2 2p^1$.

Since the 2p shell is degenerate, the electron in the last shell may be placed in any of these orbitals. Also, the wave function is rather a probabilistic representation of the reality – must include all possibilities on equal footing.

As result, there are three Slater Determinants possible for this system.

The Wave function is a linear combination between the three Slater Determinants:

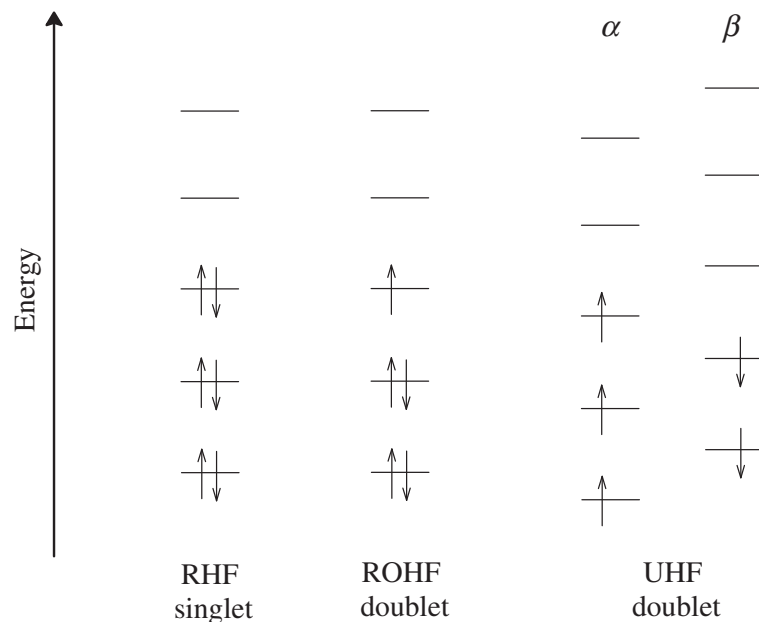
$$\psi = \frac{1}{\sqrt{3}} \left[|\varphi_{1s}^2 \varphi_{2s}^2 \varphi_{2px}^1\rangle + |\varphi_{1s}^2 \varphi_{2s}^2 \varphi_{2py}^1\rangle + |\varphi_{1s}^2 \varphi_{2s}^2 \varphi_{2pz}^1\rangle \right]$$

Given that HF method optimizes molecular orbitals of ONLY ONE Slater Determinant -- it is NOT adequate for this problem. In order to describe qualitatively correct this solution – the multi-configurational self-consistent field method (CASSCF) must be employed for the case of free B atom.

This case is a typical case of “*static electron correlation*” problem.

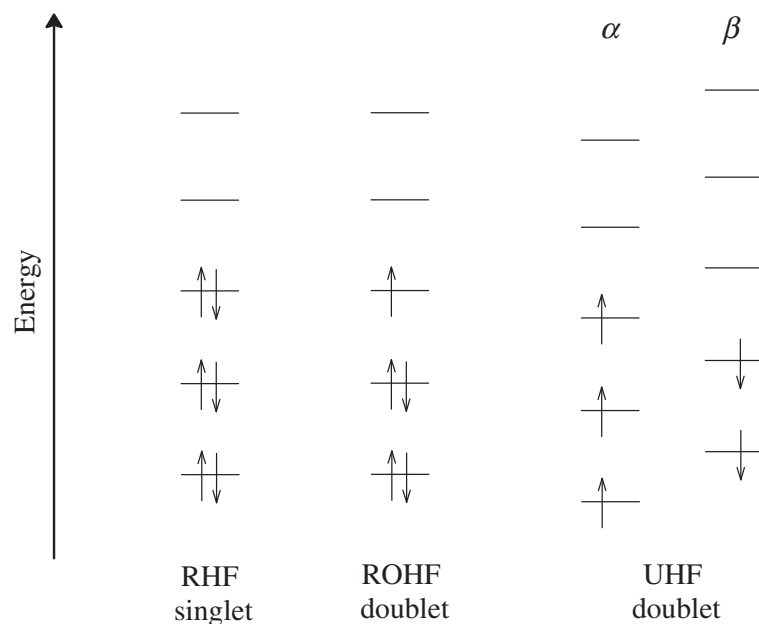
Restricted and Unrestricted SCF

- The Hamiltonian used in Hartree-Fock theory does not explicitly mention spin, and therefore, has a symmetry with respect to spin – a spin-flip does not change the energy. If we enforce that symmetry (demanding that the up and the down spin orbitals share the space part), the version of the Hartree-Fock (and more generally SCF) method that results is called *restricted*.
- In open-shell systems, the *unrestricted* version is preferred, where the two space parts are allowed to differ.



Restricted and Unrestricted SCF

- Restricted SCF is often faster (fewer integrals to evaluate), but fails to reproduce certain bond breaking processes. Unrestricted SCF treats bond breaking correctly (although for the wrong reasons), but can suffer from **spin contamination** – the resulting wavefunction is often not an eigenstate of the total spin operator.



Spin Contamination in Unrestricted SCF

- Unrestricted SCF methods produce wavefunctions that are not eigenfunctions of the total spin operator. This leads to distortions in computed magnetic properties and is best avoided.

$$\langle \hat{S}^2 \rangle_{UHF} \geq \langle \hat{S}^2 \rangle_{RHF} = S(S+1) = \left(\frac{n_\alpha - n_\beta}{2} \right) \left(\frac{n_\alpha - n_\beta}{2} + 1 \right)$$

The three common approaches to dealing with spin contamination are:

- The “**annihilation**” method projects out the terms with incorrect total spin at the end of the SCF procedure.
- The “**projection**” method projects out the terms with incorrect total spin within the SCF loop.
- The “**spin constraint**” method introduces the deviation from the desired total spin as a Lagrange term in a constrained energy optimization problem.
- Most packages print the expectation value of the total spin at the end of the SCF procedure. Large spin contamination (particularly in Hartree-Fock calculations) is usually an indication that a single determinant is a very poor approximation to the ground state – MCSCF approaches should be considered.

Basis Set approximation

- *Molecular orbitals* (MOs) are functions describing the motion of one single electron in space
- For small molecules, MOs may be directly expanded on a *grid of points* → *Numerical quantum chemistry methods*. These methods are:
 - extremely easy to parallelize → efficient usage of multi-core CPUs and GPUs
 - the size of the wave function becomes easily too large and difficult to handle
- For medium and large molecules: a functional representation of MOs is preferable, since it gives a more compact representation. As example, see the MOs of H₂ molecule:

$$\varphi_g = \frac{1}{\sqrt{2(1+S_{12})}}(1s_1 + 1s_2)$$

$$\varphi_u = \frac{1}{\sqrt{2(1-S_{12})}}(1s_1 - 1s_2)$$

Here, $1s_1$ and $1s_2$ are *atomic orbitals* of individual H atoms.

MO = LCAO

Molecular orbitals are written as a linear combination of atomic orbitals:

$$\varphi_i = \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha}$$

Basis Set approximation

Molecular orbital is written as a linear combination of atomic orbitals

$$\varphi_i = \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha}$$

$c_{i\alpha}$ —expansion coefficient of basis function α in the molecule orbital i

χ_{α} —basis function α

φ_i —molecular orbital i

Hartree-Fock equation $\mathbf{F}_i \varphi_i = \varepsilon_i \varphi_i$ can be written as:

$$\mathbf{F}_i \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha} = \varepsilon_i \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha}$$

Multiplying both sides by χ_{β}^* and integrate we have:

$$\begin{aligned} \chi_{\beta}^* \mathbf{F}_i \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha} &= \chi_{\beta}^* \varepsilon_i \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha} \\ \sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \mathbf{F} | \chi_{\alpha} \rangle &= \sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \chi_{\alpha} \rangle \varepsilon_i \end{aligned}$$

Basis Set approximation

$$\sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \mathbf{F} | \chi_{\alpha} \rangle = \sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \chi_{\alpha} \rangle \varepsilon_i$$

or, passing everything into a matrix form:

$$\langle \chi_{\beta} | \mathbf{F} | \chi_{\alpha} \rangle = F_{\alpha\beta}$$

$$\langle \chi_{\beta} | \chi_{\alpha} \rangle = S_{\alpha\beta}$$

$F_{\alpha\beta}$ = Fock matrix in atomic orbital basis set

$S_{\alpha\beta}$ - overlap matrix of the original basis set

Hartree-Fock equations in matrix form:

$$\mathbf{FC} = \mathbf{SC}\varepsilon$$

$\mathbf{C}$ = a column vector of the coefficient matrix

ε = a diagonal matrix, containing *orbital energies*.

Basis Set approximation

$$\sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \mathbf{F} | \chi_{\alpha} \rangle = \sum_{\alpha}^{N_{basis}} c_{i\alpha} \langle \chi_{\beta} | \chi_{\alpha} \rangle \varepsilon_i$$

or, passing everything into a matrix form:

$$\langle \chi_{\beta} | \mathbf{F} | \chi_{\alpha} \rangle = F_{\beta\alpha}$$

$$\langle \chi_{\beta} | \chi_{\alpha} \rangle = S_{\beta\alpha}$$

$F_{\alpha\beta}$ = Fock matrix in atomic orbital basis set

$S_{\alpha\beta}$ - overlap matrix of the original basis set

Hartree-Fock equations in matrix form:

$$\mathbf{FC} = \mathbf{SC}\varepsilon$$

$\mathbf{C}$ = a column vector of the coefficient matrix

ε = a diagonal matrix, containing *orbital energies*.

Basis Set approximation

$$F_{\alpha\beta} = \langle \chi_\alpha | \mathbf{h} | \chi_\beta \rangle + \sum_{j=1}^{N_{elec}} \langle \chi_\alpha | \mathbf{J}_j - \mathbf{K}_j | \chi_\beta \rangle = \langle \chi_\alpha | \mathbf{h} | \chi_\beta \rangle + \sum_{j=1}^{N_{elec}} (\langle \chi_\alpha \varphi_j | \mathbf{g} | \chi_\beta \varphi_j \rangle - \langle \chi_\alpha \varphi_j | \mathbf{g} | \varphi_j \chi_\beta \rangle)$$

$$= \langle \chi_\alpha | \mathbf{h} | \chi_\beta \rangle + \sum_{j=1}^{N_{elec}} \sum_{\gamma}^{N_{basis}} \sum_{\delta}^{N_{basis}} c_{\gamma j} c_{\delta j} (\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\beta \chi_\delta \rangle - \langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\delta \chi_\beta \rangle) =$$

$$= \langle \chi_\alpha | \mathbf{h} | \chi_\beta \rangle + \sum_{\gamma, \delta}^{N_{basis}} D_{\gamma\delta} (\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\beta \chi_\delta \rangle - \langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\delta \chi_\beta \rangle)$$

$$D_{\gamma\delta} = \sum_{j=1}^{N_{elec}} c_{\gamma j} c_{\delta j} \quad \text{density matrix}$$

$$F_{\alpha\beta} = h_{\alpha\beta} + \sum_{\gamma, \delta}^{N_{basis}} G_{\alpha\beta\gamma\delta} D_{\gamma\delta}$$

$$\mathbf{F} = \mathbf{h} + \mathbf{G} \cdot \mathbf{D}$$

Basis Set approximation

The HF energy can thus be computed as:

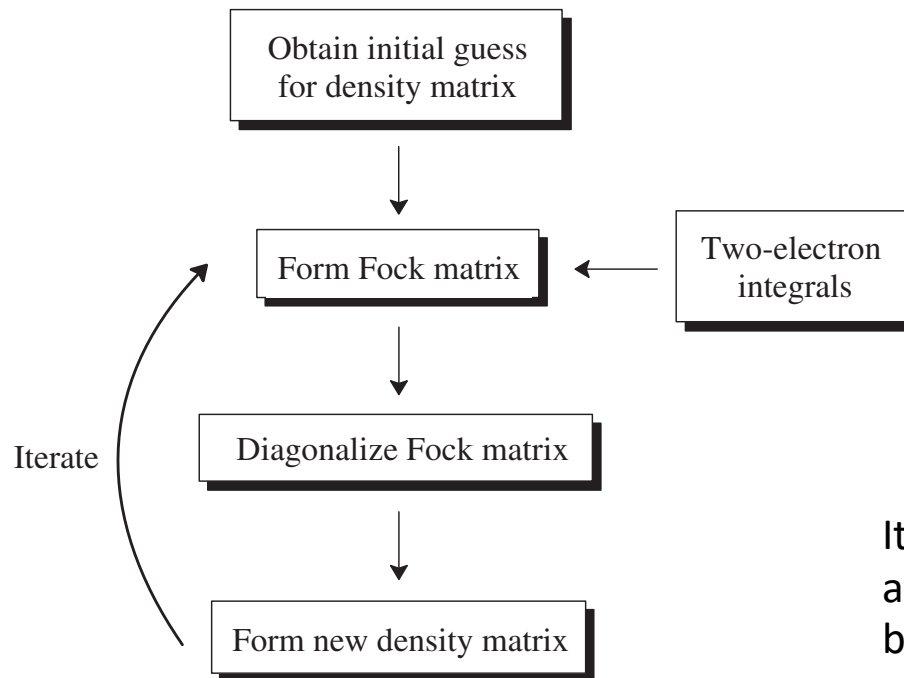
$$\begin{aligned}
 E &= \sum_i^{N_{elec}} \langle \varphi_i | h_i | \varphi_i \rangle + \frac{1}{2} \sum_{i,j}^{N_{elec}} (\langle \varphi_i \varphi_j | \mathbf{g} | \varphi_i \varphi_j \rangle - \langle \varphi_i \varphi_j | \mathbf{g} | \varphi_j \varphi_i \rangle) + V_{nn} \\
 &= \sum_i^{N_{elec}} \sum_{\alpha,\beta}^{N_{basis}} c_{\alpha i} c_{\beta i} \langle \chi_\alpha | \mathbf{h} | \chi_\beta \rangle + \frac{1}{2} \sum_{i,j}^{N_{elec}} \sum_{\alpha,\beta,\gamma,\delta}^{N_{basis}} c_{\alpha i} c_{\gamma j} c_{\beta i} c_{\delta j} (\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\beta \chi_\delta \rangle - \langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\delta \chi_\beta \rangle) + V_{nn} \\
 &= \sum_{\alpha,\beta}^{N_{basis}} D_{\alpha\beta} h_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta}^{N_{basis}} D_{\alpha\beta} D_{\gamma\delta} (\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\beta \chi_\delta \rangle - \langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\delta \chi_\beta \rangle) + V_{nn}
 \end{aligned}$$

$\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\beta \chi_\delta \rangle$ - *Coulomb integral*

$\langle \chi_\alpha \chi_\gamma | \mathbf{g} | \chi_\delta \chi_\beta \rangle$ - *Exchange integral*

SCF Procedure

$$E = \sum_{\alpha,\beta}^{N_{basis}} D_{\alpha\beta} h_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta}^{N_{basis}} D_{\alpha\beta} D_{\gamma\delta} (\langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\beta} \chi_{\delta} \rangle - \langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\delta} \chi_{\beta} \rangle) + V_{nn}$$



$$D_{\gamma\delta} = \sum_{j=1}^{N_{elec}} c_{\gamma j} c_{\delta j}$$

$$F_{\alpha\beta} = h_{\alpha\beta} + \sum_{\gamma,\delta}^{N_{basis}} G_{\alpha\beta\gamma\delta} D_{\gamma\delta}$$

Iterate until **convergence** is reached:

- energy does not change
- density matrix does not change

Variational Principle!

SCF Procedure

After convergence is reached, molecular orbitals are known, $c_{i\alpha}$ are known.

$$\varphi_i = \sum_{\alpha}^{N_{basis}} c_{i\alpha} \chi_{\alpha} \quad ; \quad D_{\gamma\delta} = \sum_{j=1}^{N_{elec}} c_{\gamma j} c_{\delta j} \quad ; \quad F_{\alpha\beta} = h_{\alpha\beta} + \sum_{\gamma,\delta}^{N_{basis}} G_{\alpha\beta\gamma\delta} D_{\gamma\delta}$$

Any unitary transformation of φ_i does not change total energy E .

- Converged orbitals for which $D_{\gamma\delta}$ is diagonal $\rightarrow$ *Natural* Orbitals
- Converged orbitals for which $F_{\alpha\beta}$ is diagonal $\rightarrow$ *Canonical* Orbitals
- Diagonal values of the $D_{\gamma\delta}$ are *occupation numbers*. for RHF, occupation numbers are 2,2,2,2,...
- Diagonal values of the $F_{\alpha\beta}$ are *orbital energies*.

Computational aspects of SCF

After convergence is reached, molecular orbitals are known, $c_{i\alpha}$ are known.

$$E = \sum_{\alpha,\beta}^{N_{basis}} D_{\alpha\beta} h_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta}^{N_{basis}} D_{\alpha\beta} D_{\gamma\delta} (\langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\beta} \chi_{\delta} \rangle - \langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\delta} \chi_{\beta} \rangle) + V_{nn}$$

$D_{\alpha\beta}$, $h_{\alpha\beta}$ are square matrices of size N_{basis} . These matrices can be kept in the active memory of the computer (RAM) and therefore their evaluation is relatively fast.

The second summation is the hardest part.

$\langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\beta} \chi_{\delta} \rangle$ - two-electron-repulsion integrals. In function of the indices $\alpha, \beta, \gamma, \delta$, a two-electron integral may be called one-, two-, three- and four-centre integral. For combinatorial reasons, the number of four-centre integrals may be very large. They cannot be kept in memory for any compound larger than benzene.

Semi-empirical methods are based on partial neglect of two-electron integrals.

Computational aspects of 2-ERIs

Conventional methods – store pre-computed integrals on disk	Direct methods – compute integrals on-the-fly
Integrals only need to be computed once	Integrals need to be computed as many times as they need to be used (typically 4-16 times).
Sequential access bandwidth – 100MB/s, random access bandwidth – 1 MB/s, latency – 1-10 ms.	Sequential throughput – 10GB/s, random throughput – 1GB/s, latency – 1-10 ns.
For very large molecules (100+ atoms) the integral arrays do not fit on disk.	Only the basis description needs to be stored, which usually fits into memory.
Data compression is available	Data compression is not necessary

Semi-direct methods are the middle ground – they are useful for post-HF methods, which can store some very expensive quantities on disk. Semi-direct methods are often much faster on the latest (SSD) storage systems

Resolution of Identity (RI) methods

(= density fitting = Cholesky decomposition methods)

Resolution of identity (RI) is an equation connecting every complete basis set to the identity operator on that space:

$$\hat{E} = \sum_i^N |\varphi_i\rangle\langle\varphi_i|$$

Using this operator, the 2-ERI can be written as:

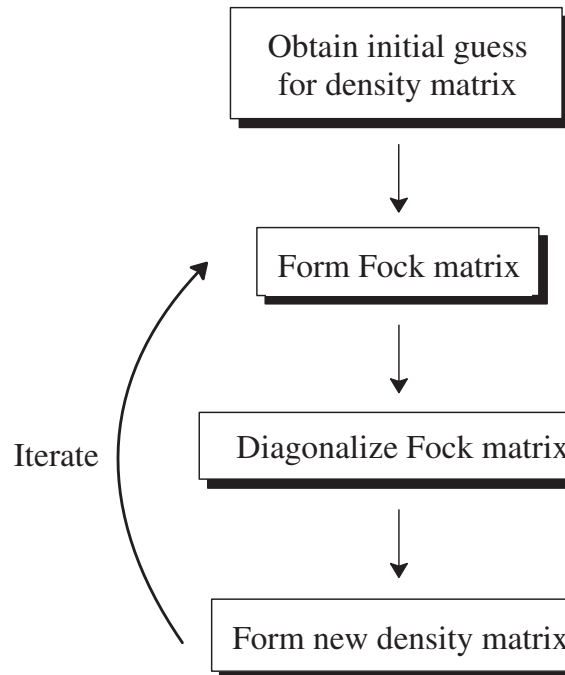
$$\langle\chi_\alpha\chi_\gamma|\mathbf{g}|\chi_\beta\chi_\delta\rangle = \sum_{P,Q}^{N_{aux\ basis}} \langle\chi_\alpha\chi_\gamma|\mathbf{g}|P\rangle (\langle P|Q\rangle)^{-1} \langle Q|\mathbf{g}|\chi_\beta\chi_\delta\rangle$$

Saving comes from the following:

- 1) $\langle\chi_\alpha\chi_\gamma|\mathbf{g}|P\rangle$ and $\langle Q|\mathbf{g}|\chi_\beta\chi_\delta\rangle$ are 3-dimensional matrices. P, Q are called *fitting basis set*, are finite and *much smaller* than the product of two original basis functions N_{basis}^2 (as it is for the product $\chi_\beta\chi_\delta$).
- 2) The matrix $\langle P|Q\rangle$ can be stored in active memory (RAM).
- 3) Original scaling of HF is N^4 . With RI – scaling is reduced to N^3 or even less, depending on the amount of RAM of the computer.
- 4) *Cholesky Decomposition* of 2-ERIs is a particular form of RI method.

Convergence issues

$$E = \sum_{\alpha,\beta}^{N_{basis}} D_{\alpha\beta} h_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta}^{N_{basis}} D_{\alpha\beta} D_{\gamma\delta} (\langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\beta} \chi_{\delta} \rangle - \langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\delta} \chi_{\beta} \rangle) + V_{nn}$$



$$D_{\gamma\delta} = \sum_{j=1}^{N_{elec}} c_{\gamma j} c_{\delta j}$$

$$F_{\alpha\beta} = h_{\alpha\beta} + \sum_{\gamma,\delta}^{N_{basis}} G_{\alpha\beta\gamma\delta} D_{\gamma\delta}$$

Iterate until convergence is reached:

- energy does not change
- density matrix does not change

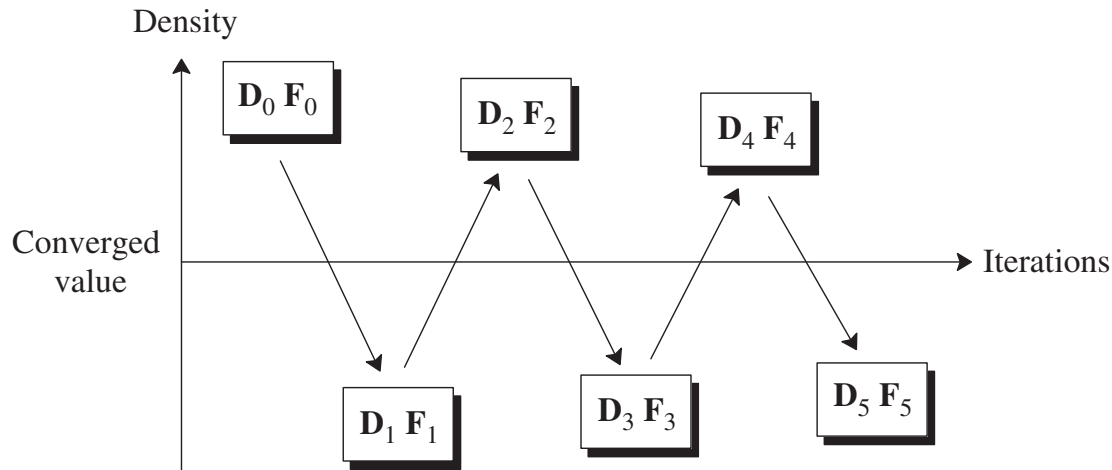
In this formulation the algorithm has *poor convergence*. A very large number of iterations is needed, and in many cases, the convergence may never be reached.

Convergence acceleration tricks

The following techniques were developed over the years in order to increase convergence rate.

1) Extrapolation: refer to extrapolate the Fock matrices from previous iterations to generate a better Fock matrix for the current iteration. Usually the Fock matrices from the last three-five iterations are used.

2) Damping: $D'_{n+1} = \omega D_n + (1 - \omega) D_{n+1}$, ω – damping factor

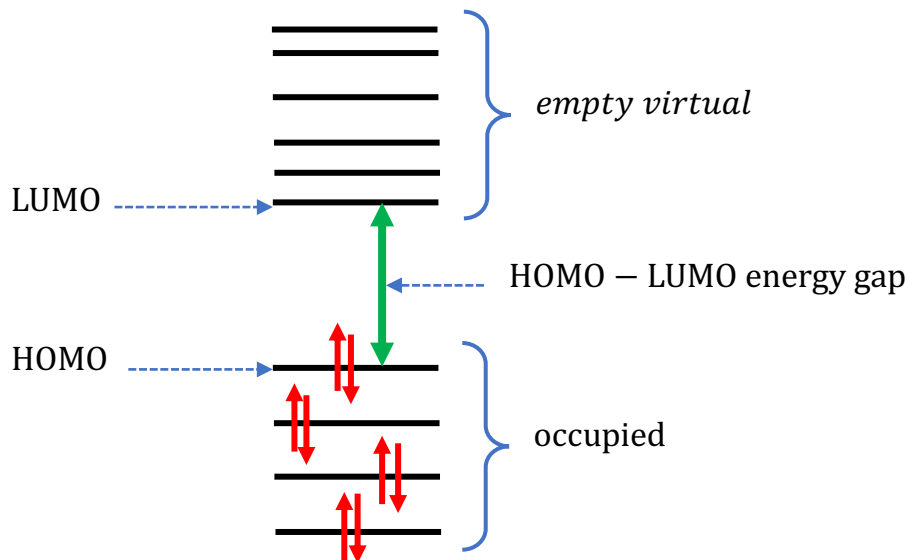


Convergence acceleration tricks

3) Level shifting: The method consists in artificially increasing the HOMO-LUMO gap, which causes immediately a *smaller mixing between occupied and empty orbitals at the next iteration*.

At full SCF convergence → the *mixing between occupied and empty orbitals* = zero (0).

By level shifting, convergence becomes guaranteed, but it may also converge to local minima.



Convergence acceleration tricks

4) Direct inversion of Iterative Subspace (DIIS):

For the iteration $n+1$, we construct the Fock matrix as a linear combination between ALL previous (n) Fock matrices:

$$F'_{n+1} = \sum_i^n c_i F_i$$

The coefficients c_i are found by minimizing the following “error” function:

$$f(\mathbf{c}) = \text{Trace} \left((\mathbf{FDS} - \mathbf{SDF})^2 \right)$$

under the condition that $\sum_i^n c_i = 1$.

$\mathbf{F}$, $\mathbf{D}$, $\mathbf{S}$ – are the Fock matrix, Density matrix and the Overlap matrix, respectively.

4) Newton-Raphson (NR), Conjugated Gradients (CG), etc. methods → they make use of the orbital Gradient (1st derivative) and orbital Hessian (2nd derivative) to predict the best Fock matrix. Convergence is good near the minimum. Hessian is expensive to calculate. NR or CG methods could be used if DIIS fails.

Effects of Symmetry

If the molecule has symmetry (e.g. square planar, inversion, octahedral, trigonal, etc.), then one could make use of this symmetry to speed-up the calculation.

- 1) Use the respective point group symmetry operations to generate a *symmetry adapted linear combination of atomic orbitals* (SALCs) as an initial guess of MOs.
- 2) Fock matrix $F_{\alpha\beta}$, core Hamiltonian matrix $h_{\alpha\beta}$, bielectronic integrals $\langle\chi_\alpha\chi_\gamma|\mathbf{g}|\chi_\beta\chi_\delta\rangle$, Density matrix $D_{\alpha\beta}$, become **block-diagonal**. An example of the block-diagonal matrix is given below:

	Irrep 1	Irrep 2	Irrep 3	Irrep 4
Irrep 1		0	0	0
Irrep 2	0		0	0
Irrep 3	0	0		0
Irrep 4	0	0	0	

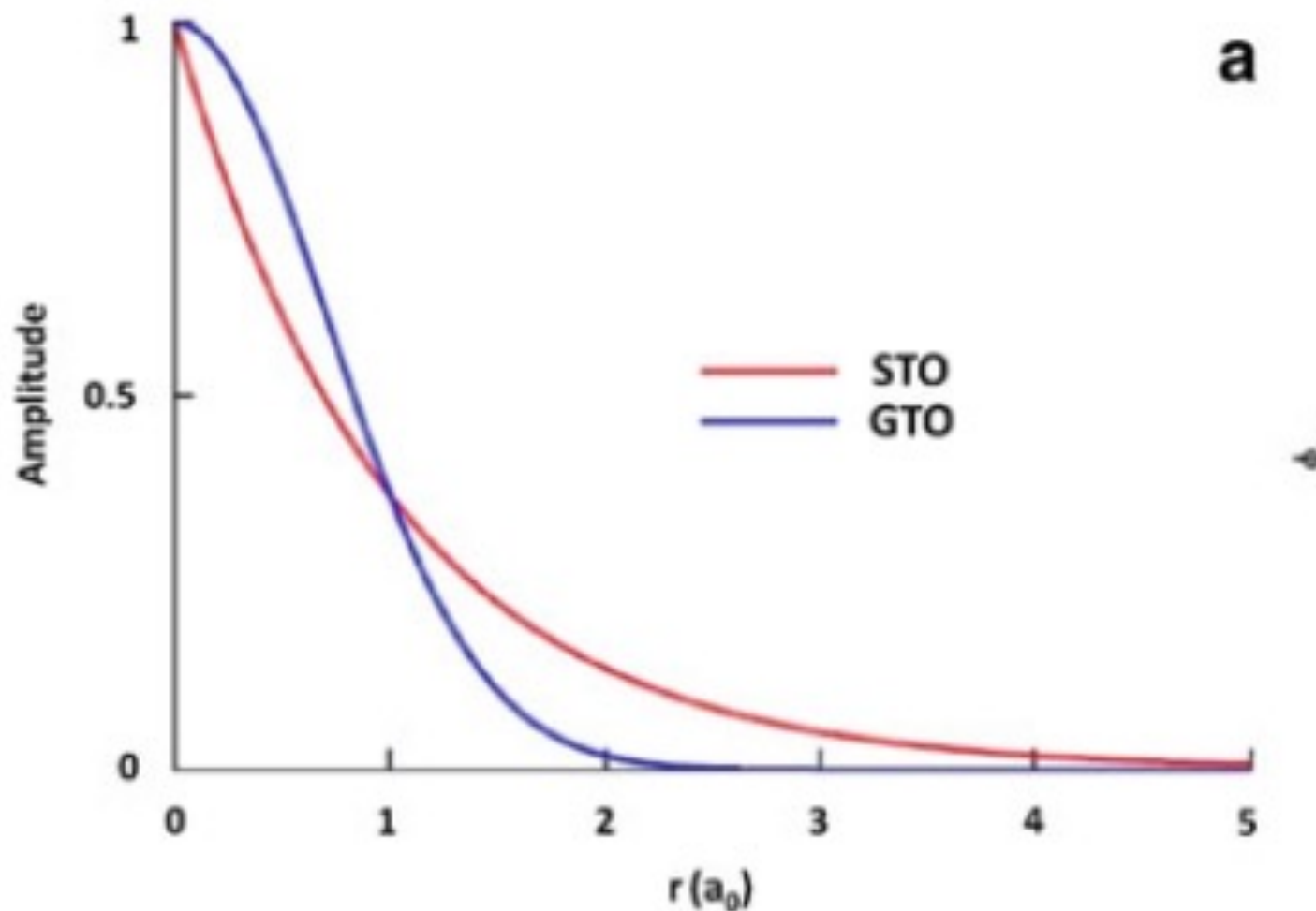
- Storing only the non-zero parts → significant space and memory saving
- Multiplication of **block-diagonal** is also much faster.
- Convergence is also enhanced.

A few words about atomic basis sets

Slater-Type Orbitals (STO's)

$$\phi_{abc}^{\text{STO}}(x, y, z) = N x^a y^b z^c e^{-\zeta r}$$

- N is a normalization constant
- a, b, and c determine the angular momentum, i.e.
 $L=a+b+c$
- ζ is the orbital exponent. It determines the size of the orbital.
- STO exhibits the correct short- and long-range behavior.
- Resembles H-like orbitals for 1s
- **Difficult to integrate for polyatomics**



Comparison of Slater- and Gaussian-type orbitals is shown schematically. The orbital exponent for both the orbitals is 1.0. Slater type orbital can be “fitted” using several Gaussian functions, with different exponents and radii.

Gaussian Basis Sets

Property	Benefits
A product of two Gaussian functions is another Gaussian at their centre of gravity.	<i>All two-electron integrals are analytical.</i> Massive time saving compared to numerical integration.
Gaussians fall off very steeply with distance.	Distance based cut-offs further improve performance.

Property	Drawbacks
Gaussians fall off very steeply with distance.	Incorrect asymptotic behaviour at large distances.
Gaussians have zero derivative at the origin.	Incorrect asymptotic behaviour at small distances.
Gaussians forming the basis are centred at the nuclei.	Basis set superposition error.

Atom-centred spherical coordinates are usually used in atomic basis sets:

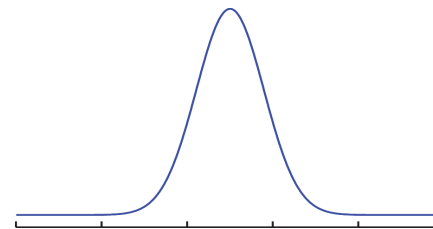
$$G_{n,l,m}(r, \theta, \varphi) = Y_{l,m}(\theta, \varphi) e^{-r^2/b_n^2}$$

spherical harmonic
Gaussian

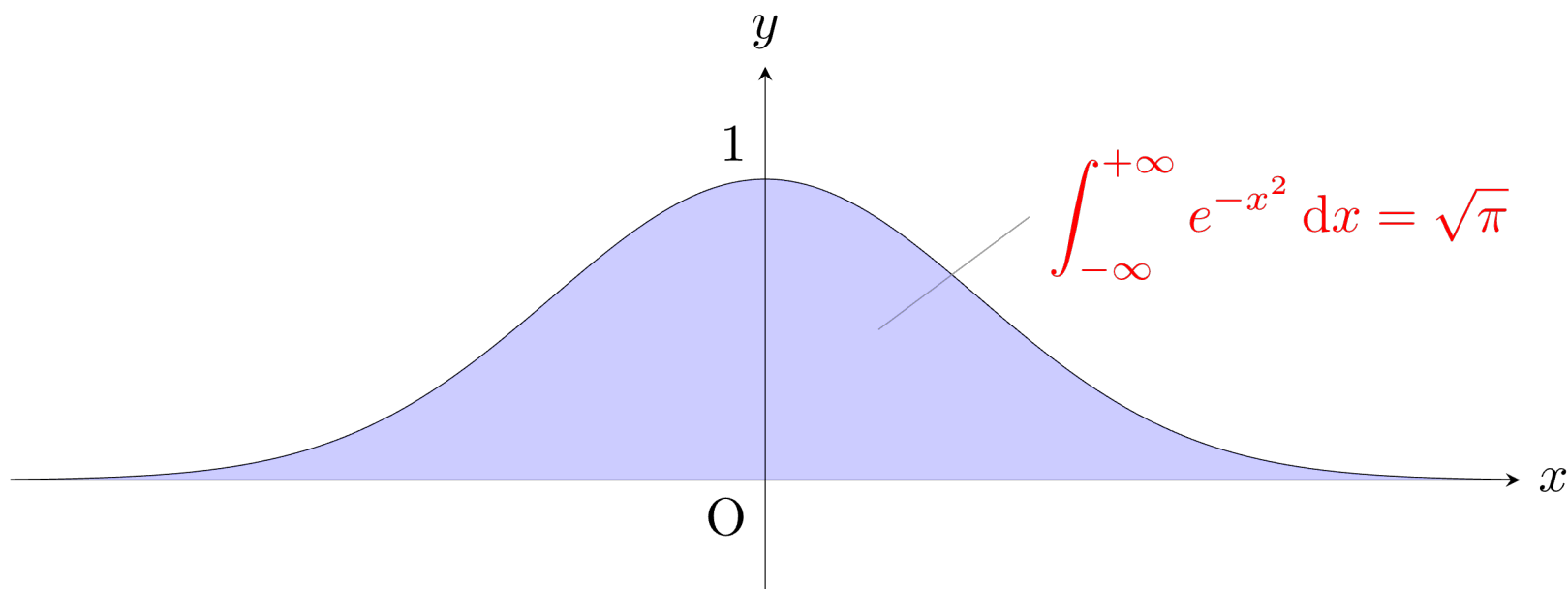
Linear combinations of Gaussians (called contractions) are often used in the radial part:

$$G_{n,l,m}(r, \theta, \varphi) = Y_{l,m}(\theta, \varphi) \sum_i a_{ni} e^{-r^2/b_{ni}^2}$$

contraction coefficients



Gaussian basis functions allow analytic integration

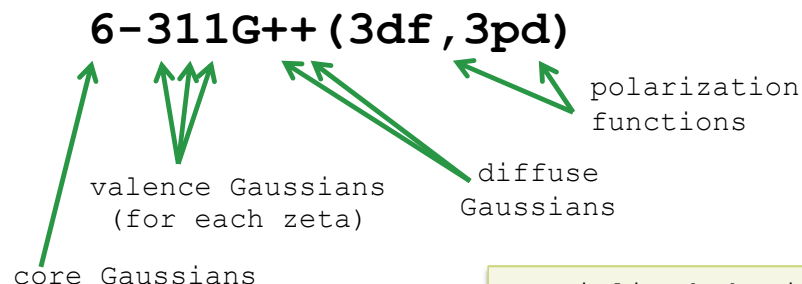


Gaussian Basis Sets

Property	Explanation
Core functions	<i>Basis functions intended to provide the description of the core electrons of an atom. Core electrons (e.g. 1s for carbon) do not participate in chemical bonding.</i>
Valence functions	<i>Basis functions intended to provide the description of the valence electrons of an atom that are involved in chemical bonding. Difference to core is often notional.</i>
Zeta number	<i>Number of independent sets of radial basis functions in the valence part of the basis (usually 2-4). They are introduced to improve basis flexibility.</i>
Polarization functions	<i>Additional functions with a higher angular momentum number, introduced to allow atomic orbitals to be shifted ("polarized") relative to their nucleus.</i>
Diffuse functions	<i>Additional functions with very broad profile, introduced to improve description of the regions that are far away from the nuclei, usually used for anions.</i>
Tight functions	<i>Additional functions concentrated tightly around their nuclei to improve the description of the nuclear region for NMR and ESR interaction calculations.</i>
Effective core potentials	<i>Effective Coulomb interaction potentials designed to replace inner orbitals of heavy atoms that are not well described by non-relativistic methods.</i>
Rydberg functions	<i>Additional extremely diffuse basis functions required for accurate description of Rydberg states.</i>

Gaussian Basis Sets

Pople's basis sets:



Correlation-consistent basis sets:

aug-cc-pVTZ

"correlation consistent
polarized valence triple-
zeta basis set augmented
with diffuse functions"

What to look for:

Specialized basis sets exist for almost any specific purpose. Check the literature for any case-specific basis sets before using a general purpose one.

Zeta number	Polarization functions	Diffuse functions	Tight functions	Effective core potentials	Rydberg functions
The more, the better/slower, usually 2-3.	The more, the better/slower, Usually one or two $l+1$ sets.	The more, the better/slower, usually a single set for anions.	Only used for computing HFCs and J -couplings.	Used to approximate inner shells of heavy elements	Only used to describe Rydberg states

Bigger and more flexible basis sets provide more accurate matrix representations for the original differential equations and therefore yield more accurate results.

Exact form of the wave function

$$\Psi = a_0 \psi_{HF} + \underbrace{\sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r}_{\text{single excitations}} + \underbrace{\sum_{\substack{i,j, \\ i < j}}^{N_{occ}} \sum_{\substack{r,s, \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs}}_{\text{double excitations}} + \underbrace{\sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst}}_{\text{triple excitations}} + \dots$$

ψ_i^r – a Slater determinant in which the electron from the orbital i is excited towards orbital r

a_i^r – expansion coefficient of this Slater determinant into the full wave function

ψ_{ij}^{rs} – a Slater determinant in which two electrons are excited

...

The *exact wave function* includes ALL possible excited Slater determinants. It can only be computed for very-very small molecules and light atoms, in a small-medium basis set.

It is called sometimes Full Configuration Interaction (FCI) wave function.

Exact form of the wave function

$$\Psi = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j, \\ i < j}}^{N_{occ}} \sum_{\substack{r,s, \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l, \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t, \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

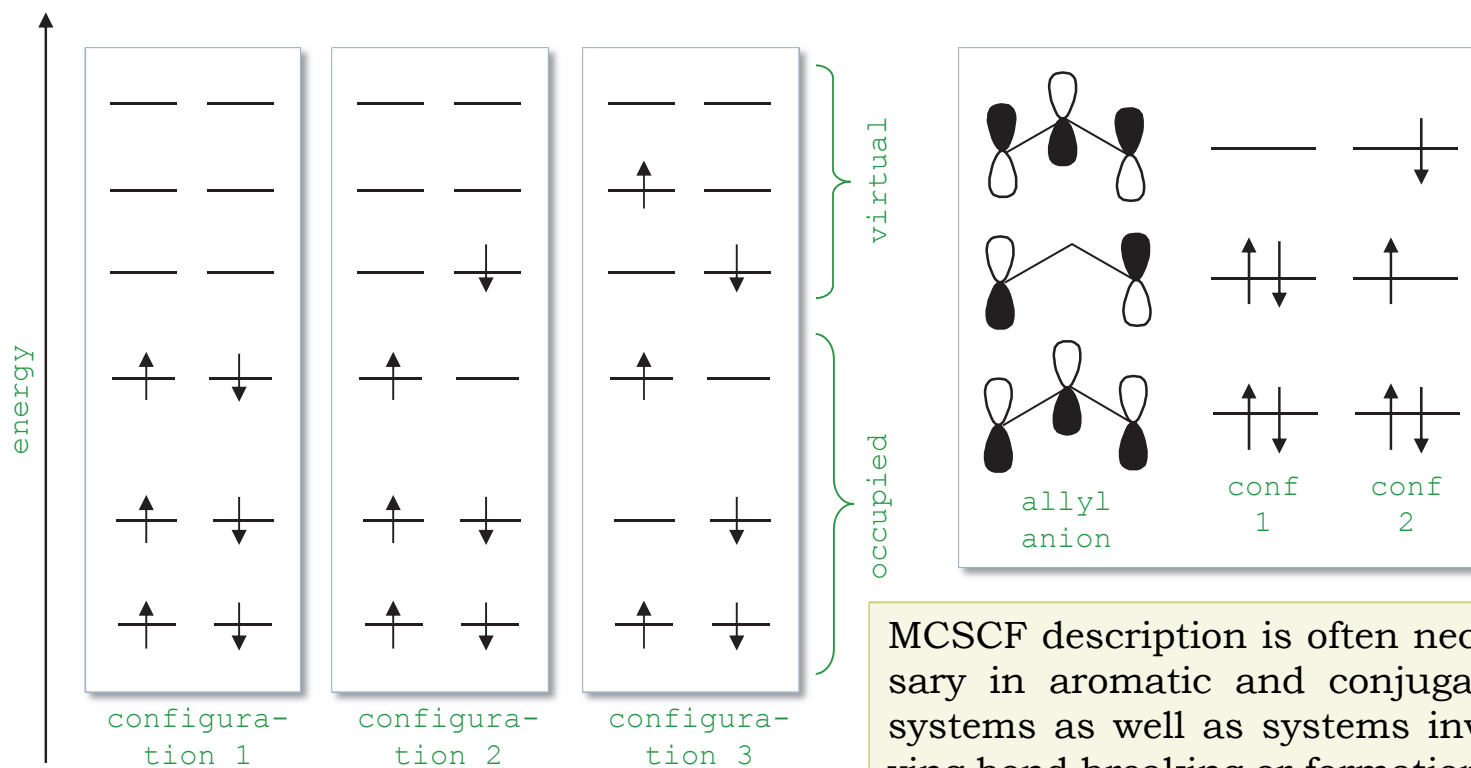
By adding excited Slater determinants into the HF wave function, the electrons start to feel each other – and avoid each other (increasing the inter-electronic distance), accounting for the dynamical correlation.

Two main groups of methods are distinguishable:

- 1) Single-reference methods (MP2, MP3, MP4, CCSD, CCSDT, CISD, CISDT, etc.)
- 2) Multi-reference methods (CASPT2, NEVPT2, etc., MRCI, MRCC, CASSCF, MCSCF, RASSCF, etc.)

Multiconfigurational SCF method

If a single Slater determinant gives a poor description of the system, multiple determinants (differing in the electron arrangement in the orbitals) may be used, with weights adjusted using the *variational principle*.



Configuration Interaction (CI)

We can take MCSCF to the limit and include all excited determinants into the calculation. This is extremely expensive (for combinatorial reasons), and some kind of truncation procedure is often required.

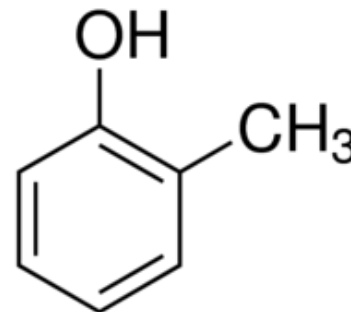
Method	Configurations	Scaling with basis size
CID	all doubly excited determinants	$O(n^5)$
CISD	all singly and doubly excited determinants	$O(n^6)$
CISDT	all singly, doubly and triply excited determinants	$O(n^8)$
full CI	all configurations	$O(n!)$

The limit where full CI is used for a subset of all orbitals is known as CASSCF (complete active space SCF). The choice of the orbitals is not a simple matter.

$$\Psi = a_0 \psi_{HF} + \sum_i^{N_{occ}} \sum_r^{N_{virt}} a_i^r \psi_i^r + \sum_{\substack{i,j \\ i < j}}^{N_{occ}} \sum_{\substack{r,s \\ s < r}}^{N_{virt}} a_{ij}^{rs} \psi_{ij}^{rs} + \sum_{\substack{i,j,l \\ i < j, \\ j < l}}^{N_{occ}} \sum_{\substack{r,s,t \\ s < r, \\ s < t}}^{N_{virt}} a_{ijl}^{rst} \psi_{ijl}^{rst} + \dots$$

Due to the astronomical number of configurations, full CI is only possible for very small (diatomic) systems. Fortunately, full CI is rarely necessary, as CISD tends to be sufficiently accurate for most practical purposes.

Practical Exercise



- 1) Open GAUSSVIEW either on your laptop or remotely on HPC
- 2) Draw a medium-sized molecule, e.g. **o-Cresol**
- 3) Setup a Gaussian calculation using the following computational options:
 - optimise molecular structure using **tight** convergence
 - use **spin-unrestricted DFT**, using PBE density functional
 - use **ultrafine** grids for integrals and DFT density integration
 - compute vibrational frequencies
 - save the Gaussian JOB on your computer (or in your \$HOME on HPC)
 - prepare the job submission script (using the example from CANVAS).
 - Transfer the Gaussian input and the job submission script to HPC
- 4) Run the calculation as a job on HPC.
- 5) After the calculation is successfully done, transfer the obtained **checkpoint file** and open it using GaussView. Inspect normal modes. Are all frequencies real (positive)? Why is this important? Was the optimisation successful?
- 6) Visualise HOMO and LUMO of the final calculation.
- 7) Inspect the calculated atomic charges (Mulliken). Which atom is the most positive charged and which one is the most negative charged?
- 8) Repeat the entire calculation using the **RevTPSS** density functional and using a **spin-restricted** formalism. What are the numerical differences of the vibrational frequencies obtained with the two calculations?
- 9) How can one assess the **structural differences** between the two optimised structures? Which one is “more” correct / reliable?